

Machine learning-predicted insulin resistance is a risk factor for 12 types of cancer

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This file contains all reviewer reports in order by version, followed by all author rebuttals in order by version.

Version 0:

Reviewer comments:

Reviewer #1

(Remarks to the Author)

Review of "Machine learning-predicted insulin resistance is a risk factor for 12 site-specific Cancers" by CL Lee et al.

The paper is generally clear and the methods are solid. I have a few comment though.

Main comments:

- All survival analyses: to start follow-up time at enrollment into the UK Biobank is not appropriate. That is not a natural starting point. The time scale of the analysis should be age and follow-up should start at age at enrollment and end at age at event or censoring.
- That is particularly important for the Kaplan Meier survival curves (e.g. Figure 2) as otherwise age, a very strong risk factor for all cancers and cardiovascular outcomes, is not accommodated at all in the curves. However, the same comment pertains to the Cox regression models, as using age as the scale of analysis adjusts for age the best way possible. Including age as a covariate is typically not sufficient.
- How is diabetes defined in the Cox regression and Kaplan Meier analysis? Are diabetes type 1 and 2 combined? I assume diabetes was observed, not predicted, is that correct?
- If yes, as date of diabetes is not known precisely, how was it used when plotting the curves? When starting individuals at baseline one must not use future information on diabetes status to stratify. One can only use baseline diabetes status for the stratification. Please clarify what was done. In the Cox model one could incorporate diabetes as a time varying exposure.
- When looking at individual cancer entities a multiple testing adjustment should be considered. renal pelvis cancer, leukemia, and renal pelvis cancer will not be significantly associated with insulin resistance after multiple testing correction.
- Page 10, lines 175-177: how much were the associations with the predicted insulin resistance score attenuated after adjusting for BMI? please comment.
- Lines 179-181: All lung cancer findings need to be adjusted or stratified by smoking status initially. They are otherwise extremely difficult to interpret.
- Lines 185-192: How was never/previous smoking defined? Are these categories in the UK biobank?
- Figure 3: the BMI adjusted and unadjusted analyses should be conducted in the same individuals, the unadjusted analysis was done in 94 782 people and the adjusted one in 20938. The differences in effect sizes could be due to differences in the 2 populations.
- Analyses should also be stratified by gender numbers permitting.

Minor point: was the breast cancer analysis restricted to women?

Reviewer #2

(Remarks to the Author)

In this article the authors use ML to find predictors of insulin resistance and evaluate its link with cardiovascular disease, diabetes and cancer. The authors state that there is limited epidemiological evidence linking insulin resistance and cancer, as the premise of developing their machine learning model. They developed their ML model using NHANES and the Taiwan MJ cohort. They propose using the ML model in clinical practice to identify those at risk diabetes, CV disease and cancer.

Their results support their conclusions that insulin resistance is associated with an increased risk of cancer.

A major concern about the data presented in the manuscript is the novelty, and significance of their results with regard to advancing the field. The authors use parameters that are well-known to be associated with insulin resistance, as described by Gerald Reaven in 1988. The metabolic syndrome is defined by three of five components hyperglycemia, dyslipidemia (elevated TGs and low HDL) and abdominal obesity, or obesity by BMI, in addition to elevated blood pressure as its defining criteria. The metabolic syndrome is a well-known syndrome of insulin resistance, carries with it an increased risk of diabetes, cardiovascular disease and cancer (in addition to other conditions such as fatty liver disease, and polycystic ovarian syndrome). The metabolic syndrome is easy to define by clinical criteria, as described in 2009.

<https://www.ahajournals.org/doi/10.1161/CIRCULATIONAHA.109.192644>

Surprisingly, nowhere in the article do the authors discuss the metabolic syndrome.

The authors state that there is a lack of epidemiology data linking insulin resistance to cancer, but there are a number of studies including the Women's Health Initiative Observational studies, where fasting insulin samples were available, that linked hyperinsulinemia to different cancers, including endometrial cancer.

<https://pubmed.ncbi.nlm.nih.gov/18172327/>

<https://pubmed.ncbi.nlm.nih.gov/18398032/>

The background literature linking the metabolic syndrome with cancer risk is missing. Below are just a few links to studies, meta-analyses dating back more than 10 years, as well as newer NHANES and China population research.

<https://pubmed.ncbi.nlm.nih.gov/23640372/>

<https://pubmed.ncbi.nlm.nih.gov/23093685/>

<https://pubmed.ncbi.nlm.nih.gov/24402120/>

<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3191378/>

<https://acsjournals.onlinelibrary.wiley.com/doi/full/10.1002/cncr.35235>

<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC9844618/>

The authors find that fasting glucose, low HDL and high TGs are predictors of cancer independent of BMI. From the above Circulation article, the metabolic syndrome has been defined for 15 years without the need for elevated waist circumference or BMI, as it is known that insulin resistance can occur in the absence of obesity. Their finding of insulin resistance as a risk factor independent of BMI is not new.

They also describe that obesity is an important factor in driving the metabolic syndrome cancer risk. Obesity is a well-known risk factor for a number of cancers.

The authors trained the model in their other manuscript on the NHANES and Taiwan populations. Did they take into account that insulin resistance frequently occurs in Asian populations at lower BMI than in European populations.

In the statistical analysis, did the authors correct for multiple comparisons? Are all comparisons between all groups statistically different in Table 1, and supplementary table 1?

In the discussion section, beginning at line 254, the authors state that the ML model could be implemented in clinical practice at low cost. Given that we already have clinical criteria to define the metabolic syndrome, that correlates with insulin resistance, and is a well-known risk factor for diabetes and cardiovascular disease, that are inexpensive and easy to measure in any clinical practice, a critical question that the authors do not address is what the advantage of their ML model is over the current metabolic syndrome definition.

In the manuscript the authors cite two mendelian randomization studies looking at MR predicted fasting insulin and cancer risk. There are a number of similar MR studies, some of which are summarized in this review article.

<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC9398112/>

Minor comments

Line 59:

o Current: "In this regard, it is concerned that insulin resistance and hyperinsulinemia..."

o Suggested: "in this regard, it is theorized that" or "In this regard, it has been proposed that" etc.

Line 61-62:

o Should be "insulin insulin-like growth factor 1 receptor"

Line 62:

o Missing article

o Current: "Mechanistic link between inflammation and cancer.."

o Suggested "The mechanistic link between inflammation and cancer.."

Line 241:

o Current: "Future researches are anticipated to explore.."

o Suggested: "Future research is anticipated to explore"

(Remarks to the Author)

In their manuscript, "Machine learning-predicted insulin resistance is a risk factor for 12 site-specific cancers" Lee et al use a previously described machine learning predicted insulin sensitivity score to examine the impact of insulin resistance upon cancer incidence, arguing that this score could have clinical utility for identifying patients who may benefit from increased screening. The question the authors are asking is very significant to the field because, as they highlight, it has been difficult to assess the impact of many of the individual features of systemic metabolism upon cancer risk. Unfortunately, the findings as presented need to be contextualized and interpreted in the context of both the obesity and cancer literature as well as the cancer metabolism/genetics literature, for as written the authors overstate the potential utility of this approach for cancer risk assessment and make claims related to potential causal relationships (e.g. sun exposure in the obese) for which no supportive data is provided, when activation of INSULIN signaling in certain mutational contexts drive senescent phenotypes (as a potential example in melanoma). By contrast a discussion of the mutational profiles of Uterine and Esophageal tumors both of which frequently mutate downstream components of the insulin signaling cascade (e.g. PI3K, PTEN) may help to explain why these observations are biologically significant. The key issue is that as described throughout their manuscript, the model relies heavily on BMI which is already a known risk factor for 13 tumor types (Lauby-Secretan B, 2016 NEJM, PMID: 27557308). This well established observation diminishes the potential utility of the model since BMI is one of its key inputs. It might be helpful to compare how BMI alone performs as a risk factor for these tumor types, and then potentially examine the impact of insulin sensitivity scores in patients with "normal" BMIs, as that population would not already be identifiable by their physicians as being at elevated risk.

The work has significant promise, but as written it again needs to be contextualized in order for readers to be able to understand its significance, which appears to be reduced from the claims being made by the authors. In many ways as currently written this feels like a first step, where they have applied their previously developed tool to a second data set, confirming its utility, but then highlight findings that need to be contextualized. These limitations in the data as presented and the need for additional context within the obesity and cancer literature result in the authors overstating the potential utility of their Machine Learning approach and lead them to overlook, or fail to emphasize some of their findings which are perhaps more novel and may warrant deeper discussion/analysis. In particular the impact of insulin sensitivity on lung cancer where they see a signal that may well be independent of obesity and its subsequent inflammation.

Specific points:

The key component missing in this manuscript is an open assessment of how their model compares to the use of BMI alone. Conceptually it would also benefit if the authors engaged in a frank discussion of how their Machine Learning approach fits within the extensive literature about how the components of systemic metabolism drive cancer risk in the tumor types they are highlighting.

The authors fail to build out the logic that underlies their observation in the differences between smokers/non-smokers at the end of the results section. It seems as if there would be multiple variables including lung cancer screening which is typically only done in smokers and could account for the differences. Inclusion of an epidemiologist and biostatistician to ensure you are not suffering from observation bias might be useful.

The claim that obesity might relate to sun exposure is not cited and would need to be backed up to remain (line 220).

The statistical methods are not robustly discussed. It would help readers interpret their findings if the authors expanded the methods to include how they are adjusting for Age, Sex, and BMI explicitly.

The discussion includes a "future directions" component that discusses further validating the model, which is not the focus of this manuscript but the previous one, so the discussion of correlating hyperinsulinemic-euglycemic clamp data with HOMA-IR, and their model seems out of place.

The key findings would significantly benefit from validation in independent data sets.

Minor points:

There are several instances where the authors seem to have missed components in their editing (eg. Line 188 where it just says reference).

Methods do not include sufficient detail to evaluate study.

Reviewer #5

(Remarks to the Author)

I co-reviewed this manuscript with one of the reviewers who provided the listed reports. This is part of the Nature Communications initiative to facilitate training in peer review and to provide appropriate recognition for Early Career Researchers who co-review manuscripts.

Reviewer #6

(Remarks to the Author)

The paper "Machine learning-predicted insulin resistance is a risk factor for 12 site-specific cancers" examines the link between insulin resistance, as predicted by a machine learning model, and the risk of certain cancers. Using a machine learning model developed with clinical markers (like BMI, FPG, cholesterol levels), the study assessed insulin resistance within the UK Biobank cohort and identified associations with 12 specific cancers, such as uterine, kidney, and colon cancer. The study found that insulin resistance heightened cancer risk, with both BMI-dependent (e.g., colon cancer) and BMI-independent effects (e.g., uterine cancer).

This approach suggests that using machine learning-predicted insulin resistance could enhance targeted cancer screening and preventive strategies, especially for high-risk individuals identified through metabolic health markers. The findings of this paper could potentially have a significant impact on public health.

There are a few concerns and suggestions:

1. It appears that the authors tested for insulin resistance across various cancers but found no significant association when considering all cancer types together. However, they identified 12 specific cancers (referred to as site-specific cancers) that individually show a connection to insulin resistance. While this finding is intriguing, it lacks generalizability. I suggest combining the data for these 12 site-specific cancers and re-testing the hypothesis. Since each cancer individually links to insulin resistance, a combined analysis might reveal a stronger overall signal, potentially supporting a broader association. If this additional experiment confirms a link, it could suggest an underlying biological connection across these cancers; if not, it may indicate that insulin resistance acts more as a confounder specific to these cancer types.
2. The paper's use of a machine learning approach for predicting insulin resistance could be strengthened by comparisons to simpler, established metrics, such as the TG/HDL ratio or TG-glucose index, to evaluate whether the model offers significant advantages over these alternatives. Additionally, while the authors mention in lines 67-68 that HOMA-IR is considered the best alternative to the hyperinsulinemic-euglycemic clamp, they do not elaborate on the reasoning behind this assertion, leaving a gap in the justification for their methodological choices.
3. The study primarily focuses on statistical associations, without exploring the biological mechanisms that might link insulin resistance to cancer. A deeper examination of these underlying mechanisms—such as insulin's role in cellular proliferation or its involvement in inflammation pathways—could strengthen the paper by connecting the statistical findings to established aspects of cancer biology.
4. While smoking is examined as an interaction effect in lung and bronchial cancers, it is not comprehensively addressed across other cancer types, which may influence results for cancers strongly linked to smoking, especially esophageal cancer. Including additional stratification analyses or adjusting for smoking across other cancer subtypes could help reduce potential confounding effects and enhance the robustness of the findings.

Version 1:

Reviewer comments:

Reviewer #2

(Remarks to the Author)

Thank you to the authors for addressing my previous comments, and editing the manuscript.

Related to the metabolic syndrome compared to the AI-IR to the metabolic syndrome to predict diabetes. As age is a component of their AI-IR score, and the risk of diabetes increases with age, it is not surprising that adding age to other metabolic syndrome criteria increases the risk of diabetes. The authors adjusted for age, sex and BMI in lines 113 to 115. Did the authors adjust for age when comparing the metabolic syndrome to the AI-IR, as metabolic syndrome criteria and age are clinical parameters available to any clinician without the need for a machine learning model to predict the risk of diabetes.

Lines 145 - that the AI-IR model is better than BMI to predict diabetes is not surprising, given that the AI-IR model includes the metabolic syndrome parameters that are known to be better predictors than BMI alone.

It is surprising however, that the authors found no association between obesity and the risk of diabetes - do the authors have any explanation for this, as the risk of diabetes is known to increase with increasing obesity.

What is the rationale for the "double positive" groups - BMI is part of the AI-IR model, so adding BMI in again on top of that is presumably artificially inflating the association. Could the authors clarify exactly what "single positive" and "double positive" means related to BMI and metabolic syndrome. It means presence or absence of metabolic syndrome and obesity as defined by BMI?

The criteria used to define the metabolic syndrome seem to be missing from the methods.

In the discussion the authors argue that the AI-IR model would alter screening for diabetes, cardiovascular disease and cancer. Could the authors elaborate on what screening guidelines would be changed? The study includes individuals from the UK Biobank between 40 and 69 years old - it is unclear how would these findings would change the current screening guidelines for diabetes and cancer in individuals in this age group.

The last sentence of the discussion says that the ML approach could help gain insight into disease pathophysiology. It is not clear how their AI-IR approach provides any new insight into disease pathophysiology.

(Remarks on code availability)

Reviewer #3

(Remarks to the Author)

The authors have been largely responsive to the concerns raised in my initial review.

(Remarks on code availability)

Reviewer #7

(Remarks to the Author)

Thank you for asking me to review this paper. It is a very interesting paper with highly pertinent results.

I do have a few comments for the authors:

I agree with the previous reviewer, the models should have age as the underlying time variable. Adjusting and stratifying by age at recruitment is not sufficient to fully account for age in the models. To ensure your models are robust, you should re-run them with age as the underlying time variable.

UKB does not have fasting glucose available, fasting samples were not collected. You have not addressed this in your paper, how could this affect the prediction of IR given the original model was training on fasting glucose levels? This should be addressed.

You have undertaken a impressive amount of analysis. However, this makes the methods very difficult to follow, and I'm not sure that your work would be replicable with the current methods section. I think you should consider rewriting this section so it is easy to follow and understand exactly what was undertaken in each analysis. Additionally, I am not convinced that the correct censor date has been applied to the non-cancer participants. The the two data fields listed for applying censor date are country of birth data fields, so I cannot see how you used these. Further, all first cancer diagnoses are present in the cancer registry, so absence indicates no cancer diagnosis, so the censor date of the cancer registry should be applied to all participants, regardless of if they were diagnosed with a cancer or not.

How model adjustments were applied is not clear - I was under the understanding from the methods that all models were adjusted for age, sex and BMI, however the section from line 232 suggests that is not the case.

Did you consider co-linearity between the various measures investigated, given that all these variables are included in the AI-IR model?

line141 - Can you test to see if there is a significant difference between AUCs?

lines 142 - 144: it is really not clear what has been done here

line 148: given this contradicts previous research, can you be confident in your results? What models did you use to undertake this analysis?

lines 153 - again, it's really unclear what you have done here, what are the 4 groups?

lines 184-193: I think only Bonferroni sig results should be reported

lines 200 - 217: I'm not convinced this section adds to the paper, it feels like analysis for analysis sake, rather than truly trying to understand the aetiology (which would be best done individually for each cancer site)

line 404: there is no non-UK area covered by UKB - can you clarify what you mean by this?

line 410: include ICD codes used.

(Remarks on code availability)

Whilst the code for the machine learning predicted insulin resistance is under patent, I do feel like the code for this analysis should be provided (that is, the code for these analysis, without the code to generate the AI-IR values) so it is transparent exactly how the analysis was undertaken.

Reviewer #8

(Remarks to the Author)

Authors seemed to have addressed Reviewer #6 comments.

(Remarks on code availability)

Version 2:

Reviewer comments:

Reviewer #7

(Remarks to the Author)

Thank you for resubmitting this manuscript. You have addressed all my previous concerns.

(Remarks on code availability)

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Point-by-point response to the editor and reviewers

Lee CL., Hiraike Y., et al. NCOMMS-24-48120-T

Reviewer #1 (Remarks to the Author): Expert in cancer epidemiology and statistics

Review of “Machine learning-predicted insulin resistance is a risk factor for 12 site-specific Cancers” by CL Lee et al.

The paper is generally clear and the methods are solid. I have a few comments though.

Response

The authors would like to thank the reviewer for positively evaluating our manuscript. To address concerns raised by this reviewer and strengthen or fine-tune our conclusion, we performed additional analyses. Regarding the effect of age on the incidence of cancer, we confirmed that the incidence of cancer increased with age. Moreover, both predicted insulin resistance (AI-IR) and diabetes were associated with a higher incidence of cancer across all age group, indicating that AI-IR is the risk factor for cancer regardless of age. We further addressed all other specific comments and revised the manuscript accordingly.

Main comments:

– All survival analyses: to start follow-up time at enrollment into the UK Biobank is not appropriate. That is not a natural starting point. The time scale of the analysis should be age and follow-up should start at age at enrollment and end at age at event or censoring.

Response

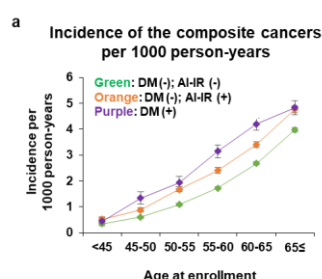
The authors feel that a Kaplan-Meier analysis starting from age at enrollment is a kind of standard analysis in the community. Having said that, we sincerely agree that age is a very strong risk factor for all cancers and cardiovascular outcomes and we need to consider the effect of age when evaluating the effect of AI-IR. To address this point, we performed multiple additional analyses. Please refer to our response to the next comment.

– That is particularly important for the Kaplan Meier survival curves (e.g. Figure 2) as otherwise age, a very strong risk factor for all cancers and cardiovascular outcomes, is not accommodated at all in the curves. However, the same comment pertains to the Cox regression models, as using age as the scale of analysis adjusts for age the best way possible. Including age as a covariate is typically not sufficient.

Response

In response to this very important comment regarding the effect of age on the incidence of cancer, we performed multiple additional analyses. First, we evaluated the incidence of the composite cancers per 1000 person-years across different age groups and showed that AI-IR were associated with a higher incidence of the composite cancers across different age groups (page 14, line 221 in the revised manuscript).

“Cancer risk is well known to increase with age^{29,30}. Indeed, when we stratified participants by age at enrollment (40 to 69 years old), the incidence of the composite cancers per 1000 person-years increased with age (Supplementary Fig. 4a). Furthermore, both AI-IR and diabetes were associated with a higher incidence of the composite cancers across all age groups (Supplementary Fig. 4a), indicating that AI-IR is the risk factor for the composite cancers across different age groups.”

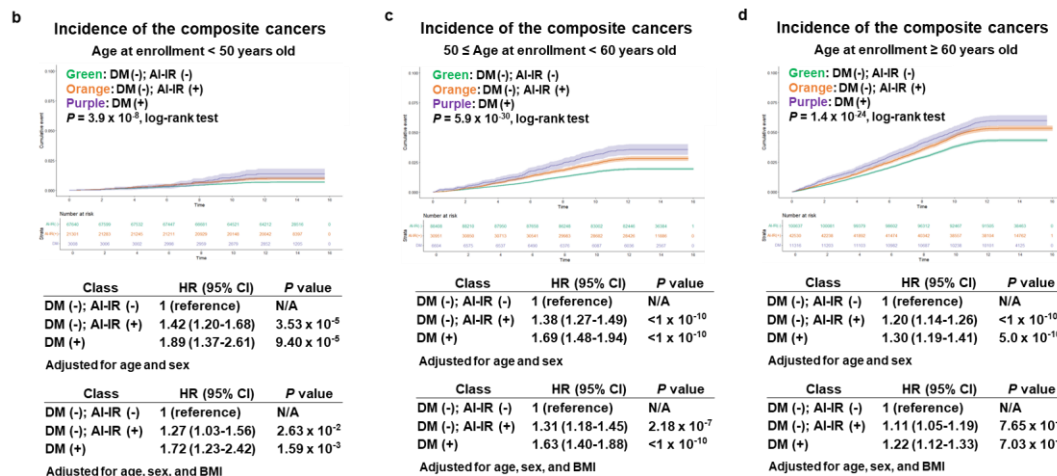


Supplementary Figure 4

Of note, we defined composite cancers by merging 10 cancer types (common to both males and females) whose risks were either significantly increased (kidney, esophagus, pancreas, colon) or nominally increased (renal pelvis, small intestine, stomach, liver and GB, colon, and bronchial and lung) by AI-IR (page 12, line 200 in the revised manuscript).

We also performed Kaplan-Meier analyses stratified by participants' age at enrollment (page 14, line 226 in the revised manuscript).

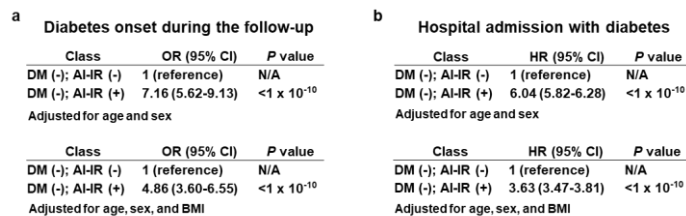
“Kaplan-Meier analyses also revealed that significant difference between participants with diabetes, those without diabetes but positive for AI-IR, and those without diabetes and negative for AI-IR on the cumulative incidence of the composite cancers are consistent across different age groups: <50 years old, 50–59 years old, and ≥60 years old at enrollment (Supplementary Fig. 4b-d).



Supplementary Figure 4

Furthermore, regarding the Cox regression models, in addition to univariate analysis, we also included age-adjusted analyses (page 7, line 112 in the revised manuscript).

“Of note, AI-IR was associated with a significantly higher risk of diabetes onset and hospital admission with diabetes after adjusting for age and sex, and even after adjusting for age, sex, and BMI, indicating that our machine learning-based prediction model is able to capture a BMI-independent effect of insulin resistance (Supplementary Fig. 1a-b).”



Supplementary Figure 1

– How is diabetes defined in the Cox regression and Kaplan Meier analysis? Are diabetes type 1 and 2 combined? I assume diabetes was observed, not predicted, is that correct?

Response

In response to this comment, we added more description into the method section (page 24, line 406 in the revised manuscript). We used observed phenotype of diabetes for the current study, and we included all subjects with diabetes for analyses.

“Observed phenotypes of diabetes was used for the current study. Individuals with HbA1c > 6.5% at enrollment and/or self-report of physician’s diagnosis were used to define diabetes. For short-term follow-up analysis in the UK Biobank, HbA1c at the

follow-up and/or self-report of physician's diagnosis were used to define new-onset of diabetes. For long-term follow-up analysis using the linkage to the NHS medical record system, diagnosis registered in NHS was used for the analysis. Diagnosis date was defined as the date of admission with diagnosis of diabetes. We included all subjects with diabetes for aforementioned analyses."

– If yes, as date of diabetes is not known precisely, how was it used when plotting the curves? When starting individuals at baseline one must not use future information on diabetes status to stratify. One can only use baseline diabetes status for the stratification. Please clarify what was done. In the Cox model one could incorporate diabetes as a time varying exposure.

Response

We used baseline status (participants without diabetes and negative for AI-IR, those without diabetes but positive for AI-IR, and those with diabetes) for stratification (page 11, line 167 in the revised manuscript).

"To investigate the effect of AI-IR on the incidence of cancer, we leveraged the linkage between the UK Biobank and the NHS medical record system and examined the incidence of cancer among participants who were cancer free at the baseline visit (N = 372,395). We compared participants without diabetes and negative for AI-IR (N = 256,685), those without diabetes but positive for AI-IR (N = 94,782), and those with diabetes (N = 20,928). Of 372,395 participants who were cancer free at baseline, 51,193 developed cancer (Supplementary Table 3)."

– When looking at individual cancer entities a multiple testing adjustment should be considered. renal pelvis cancer, leukemia, and renal pelvis cancer will not be significantly associated with insulin resistance after multiple testing correction.

Response

The authors would like to thank the reviewer for this really important comment. We performed Bonferroni correction after multiple testing in the revised manuscript (page 11, line 177 in the revised manuscript)

"However, when we looked at individual types of cancer, both AI-IR and diabetes were associated with a significantly higher risk of incidence of multiple cancer types (The results of the 25 common cancers are shown in Fig. 3, and the complete list can be found in Supplementary Table 4). We examined the incidence of 36 common cancers (both male and female), 4 cancer types specific to males, and 3 cancer types specific

to females. The Bonferroni-corrected P value for significance was $0.05/43 = 1.163 \times 10^{-3}$.

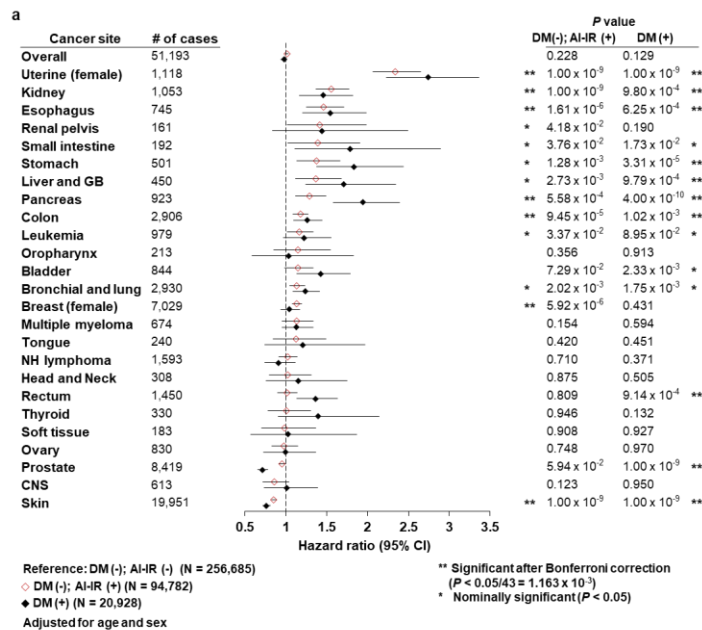


Figure 3

The results were mentioned in page 12, line 193 in the revised manuscript.

“When we considered Bonferroni correction, AI-IR was associated with a higher incidence of uterine, kidney, esophagus, pancreas, colon, and breast cancers.”

– Page 10, lines 175-177: how much were the associations with the predicted insulin resistance score attenuated after adjusting for BMI? please comment.

Response

In the revised manuscript, we defined composite cancers by merging 10 cancer types (common to both males and females) whose risks were either significantly or nominally increased by AI-IR and looked at the effect of AI-IR on the incidence of the composite cancers (page 13, line 203 in the revised manuscript). We observed that approximately 36% of the effect of AI-IR was attenuated after adjusting for BMI.

“a Kaplan-Meier plot revealed a significant difference between participants with diabetes, those without diabetes but positive for AI-IR, and those without diabetes and negative for AI-IR, on the cumulative incidence of the composite cancers (Fig. 4a). The HR adjusted for age and sex in participants without diabetes but positive for AI-IR was 1.25 (95% CI, 1.20-1.31; $P < 1 \times 10^{-11}$), and that in participants with diabetes was 1.40 (95% CI, 1.31-1.50; $P < 1 \times 10^{-11}$). When we adjusted also for BMI, The HR

in participants without diabetes but positive for AI-IR was 1.16 (95% CI, 1.10-1.22; $P = 1.2 \times 10^{-8}$), suggesting that approximately 36% of AI-IR' effect is mediated through BMI.”

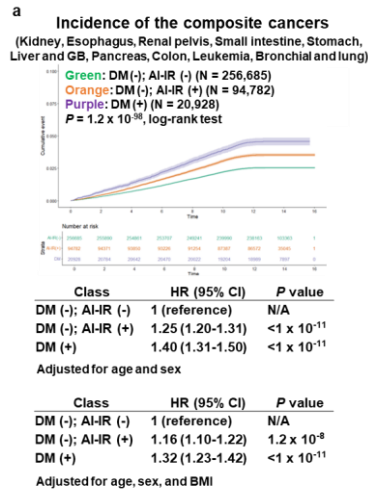


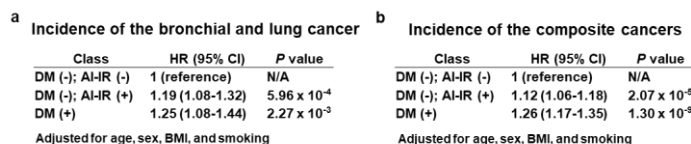
Figure 4

– Lines 179-181: All lung cancer findings need to be adjusted or stratified by smoking status initially. They are otherwise extremely difficult to interpret.

Response

In response to this comment, we additionally examined the effect of AI-IR on the incidence of bronchial and lung cancer as well as composite cancers while adjusting for smoking status as well as age, sex, and BMI, which is mentioned in page 15, line 247 in the revised manuscript.

“Lung cancer is strongly associated with smoking, the leading risk factor for the global burden of cancer³¹. We observed that the effect of AI-IR on the incidence of bronchial and lung cancer is significant even when we adjusted also for smoking status (never smoker, previous smoker, and current smoker); The same is true for the incidence of the composite cancers (Supplementary Fig. 5a-b).”



Supplementary Figure 5

– Lines 185-192: How was never/previous smoking defined? Are these categories in the UK biobank?

Response

Yes. Data field 20116 was used to define smoking status (page 25, line 425 in the revised manuscript).

– Figure 3: the BMI adjusted and unadjusted analyses should be conducted in the same individuals, the unadjusted analysis was done in 94,782 people and the adjusted one in 20,938. The differences in effect sizes could be due to differences in the 2 populations.

Response

Caption for Figure 4A (Figure 4C in the revised manuscript) was incorrect. Both BMI-unadjusted and BMI-adjusted analysis was performed in 94,782 participants without diabetes but positive for AI-IR. The BMI adjusted and unadjusted analyses were conducted in the same individuals.

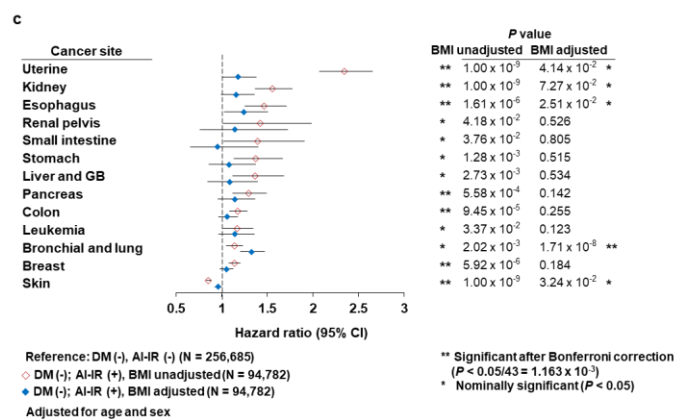


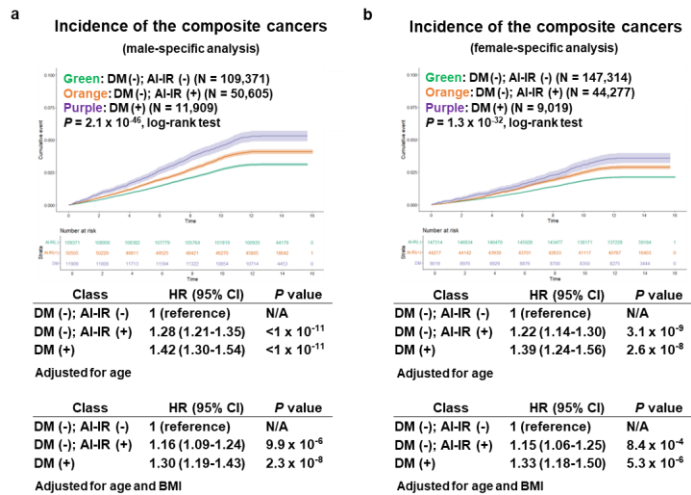
Figure 4

– Analyses should also be stratified by gender numbers permitting.

Response

In response to this comment, we additionally performed sex-specific analysis looking at the effect of AI-IR on the incidence of the composite cancers (page 13, line 210 in the revised manuscript).

“Sex-specific analysis revealed that the impact of AI-IR on the incidence of the composite cancers remained consistent in both males and females (Supplementary Fig 2a, b).”



Supplementary Figure 2

Minor point: was the breast cancer analysis restricted to women?

Response

Yes, the breast cancer analysis was restricted to women. We found that AI-IR significantly increased the incidence of both composite cancers (common to both males and females) and female-specific cancers, including uterine and breast cancers (page 13, line 212 in the revised manuscript).

“When we combined female-specific uterine and breast cancers whose risks were significantly increased by AI-IR, we observed that effect the HR adjusted for age in participants without diabetes but positive for AI-IR was 1.26 (95% CI, 1.20-1.32; $P < 1 \times 10^{-11}$), which was comparable to that in participants with diabetes (HR, 1.23; 95% CI, 1.11-1.36; $P < 3.9 \times 10^{-5}$) (Fig. 4b).”

Reviewer #2 (Remarks to the Author): Expert in insulin resistance and cancer

In this article the authors use ML to find predictors of insulin resistance and evaluate its link with cardiovascular disease, diabetes and cancer. The authors state that there is limited epidemiological evidence linking insulin resistance and cancer, as the premise of developing their machine learning model. They developed their ML model using NHANES and the Taiwan MJ cohort. They propose using the ML model in clinical practice to identify those at risk diabetes, CV disease and cancer.

Their results support their conclusions that insulin resistance is associated with an increased risk of cancer.

Response

The authors would like to thank this reviewer for constructive comments to improve our manuscript. In response to the specific comments below, we performed additional analyses and revised the manuscript to better support our conclusion. Specifically, we directly compared our prediction model, AI-IR and previously established metrics including BMI, Metabolic syndrome, TG/HDL ratio and TyG index. We conclude that AI-IR is the single best metric for predicting both diabetes and composite cancer incidence.

A major concern about the data presented in the manuscript is the novelty, and significance of their results with regard to advancing the field. The authors use parameters that are well-known to be associated with insulin resistance, as described by Gerald Reaven in 1988. The metabolic syndrome is defined by three of five components hyperglycemia, dyslipidemia (elevated TGs and low HDL) and abdominal obesity, or obesity by BMI, in addition to elevated blood pressure as its defining criteria. The metabolic syndrome is a well-known syndrome of insulin resistance, carries with it an increased risk of diabetes, cardiovascular disease and cancer (in addition to other conditions such as fatty liver disease, and polycystic ovarian syndrome). The metabolic syndrome is easy to define by clinical criteria, as described in 2009.

<https://www.ahajournals.org/doi/10.1161/CIRCULATIONAHA.109.192644>

Surprisingly, nowhere in the article do the authors discuss the metabolic syndrome.

Response

The authors would like to thank the reviewer for this important and constructive comment. In response to this comment, we refer to the metabolic syndrome in the introduction and cite the indicated article (page 5, line 72 in the revised manuscript).

“Similarly, metabolic syndrome, which is linked to visceral adiposity and its associated

insulin resistance¹¹, is also associated with a higher risk of cancer^{12–17}.”

More importantly, in response to this constructive reviewer comment, we directly compared AI-IR and Metabolic syndrome (MetS) regarding the predictive capabilities of the incidence of diabetes. We demonstrated that AI-IR is the best predictor of diabetes incidence compared to BMI, MetS, TG/HDL ratio, and TyG index (page 9, line 134 in the revised manuscript).

“To examine whether AI-IR outperforms previously established simpler metrics of obesity, metabolic syndrome, and insulin resistance, we compared the predictive capabilities of AI-IR, BMI, metabolic syndrome (MetS), TG/HDL ratio, and TyG index for the incidence of diabetes among individuals without diabetes at baseline during the follow-up period. TG/HDL ratio^{24,25} and TyG index^{26–28} has been reported as a surrogate markers of insulin resistance. Area under the curve (AUC) of the receiver operating characteristic (ROC) curve was highest for AI-IR (0.798), followed by MetS (0.748), BMI (0.721), TyG index (0.703), and Tg/HDL ratio (0.702), indicating the AI-IR exhibits the best predictive capability among these metrics (Fig. 2a).”

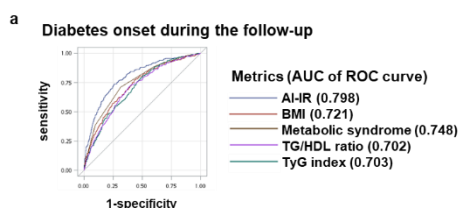


Figure 2

(page 9, line 142 in the revised manuscript)

“Next, we classified individuals without diabetes at baseline into four groups based on AI-IR status and the status of previously established metrics.”

(page 10, line 153 in the revised manuscript)

when we classified individuals without diabetes into four groups based on AI-IR status and MetS, the incidence of diabetes significantly higher ($P = 1.55 \times 10^{-3}$) for AI-IR single-positive group (OR, 6.85; 95% CI, 4.35-10.80; $P = 1.2 \times 10^{-16}$) compared to MetS single-positive group (OR, 3.14; 95% CI, 2.07-4.77; $P = 6.8 \times 10^{-8}$). Again, the incidence of diabetes was significantly higher ($P = 2.19 \times 10^{-13}$) for AI-IR and MetS double-positive group (OR, 11.71; 95% CI 8.53-16.06; $P < 1 \times 10^{-20}$) compared to MetS single positive group (Fig. 2c).

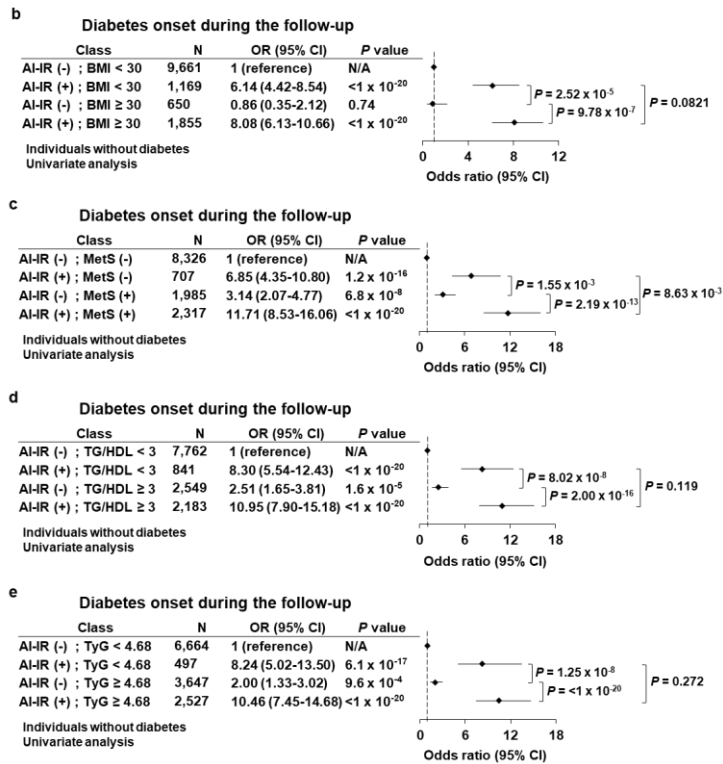


Figure 2

We also compared AI-IR and MetS regarding the prediction of the incidence of the composite cancer (page 16, line 273 in the revised manuscript).

“Finally, to examine whether AI-IR provides a better prediction of the composite cancer incidence than previously established metrics such as BMI, MetS, TG/HDL ratio, and TyG index, we classified participants who were free of diabetes and cancer at baseline into four groups based on their AI-IR status and the status of these metrics.” (page 17, line 284 in the revised manuscript)

“When considering AI-IR and MetS, the AI-IR single positive group (HR, 1.21; 95% CI 1.12-1.31; $P = 2.1 \times 10^{-6}$) and the MetS single positive group (HR, 1.19; 95% CI 1.12-1.26; $P = 1.8 \times 10^{-9}$) exhibited comparable HRs ($P = 0.649$) for the incidence of the composite cancers. However, the incidence of composite cancers was significantly higher ($P = 1.49 \times 10^{-4}$) in the AI-IR and MetS double-positive group (HR, 1.34; 95% CI 1.27-1.40; $P < 1 \times 10^{-11}$) compared to the MetS single-positive group (Fig. 5b).” (page 18, line 297 in the revised manuscript)

“Altogether, these results indicate that AI-IR enables improved cancer risk stratification compared to previously established metrics.”

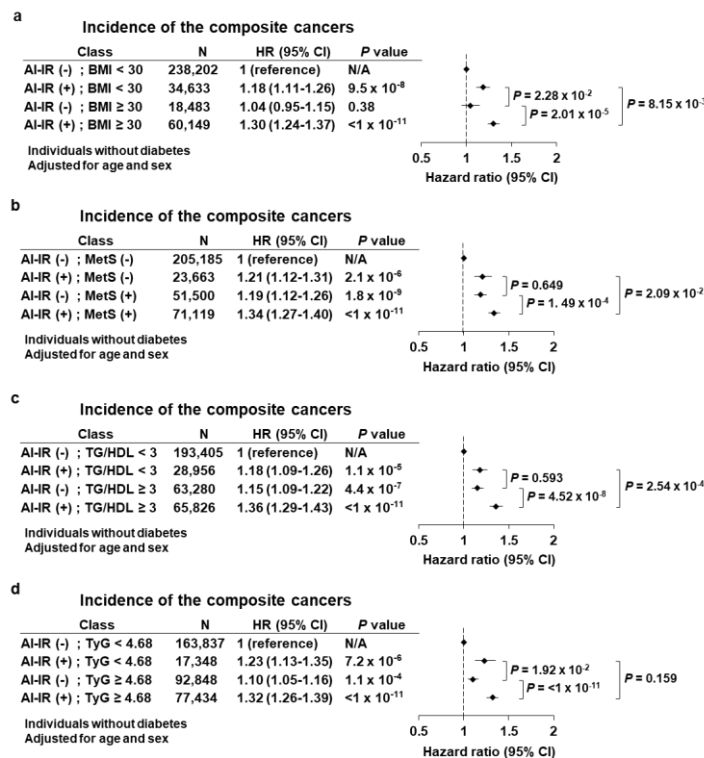


Figure 5

The authors state that there is a lack of epidemiology data linking insulin resistance to cancer, but there are a number of studies including the Women's Health Initiative Observational studies, where fasting insulin samples were available, that linked hyperinsulinemia to different cancers, including endometrial cancer.

<https://pubmed.ncbi.nlm.nih.gov/18172327/>

<https://pubmed.ncbi.nlm.nih.gov/18398032/>

Response

In response to this comment, we additionally cite indicated papers in the context of insulin resistance and cancer (page 5, line 74 in the revised manuscript)

"However, large-scale epidemiological evidence directly linking insulin resistance and cancer is limited, with only a few studies examining the incidence of endometrial and colorectal cancer using high fasting insulin as a surrogate marker for insulin resistance^{18,19}, because insulin resistance is rarely evaluated in patients, except for those being treated in specialized diabetes clinics."

The background literature linking the metabolic syndrome with cancer risk is missing. Below are just a few links to studies, meta-analyses dating back more than 10 years, as well as

newer NHANES and China population research.

<https://pubmed.ncbi.nlm.nih.gov/23640372/>

<https://pubmed.ncbi.nlm.nih.gov/23093685/>

<https://pubmed.ncbi.nlm.nih.gov/24402120/>

<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3191378/>

<https://acsjournals.onlinelibrary.wiley.com/doi/full/10.1002/cncr.35235>

<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC9844618/>

Response

In response to this comment, we refer to the metabolic syndrome in the introduction and cite the indicated articles (page 5, line 72 in the revised manuscript). We would like to appreciate this suggestion to improve our manuscript.

“Similarly, metabolic syndrome, which is linked to visceral adiposity and its associated insulin resistance¹¹, is also associated with a higher risk of cancer^{12–17}.”

The authors find that fasting glucose, low HDL and high TGs are predictors of cancer independent of BMI. From the above Circulation article, the metabolic syndrome has been defined for 15 years without the need for elevated waist circumference or BMI, as it is known that insulin resistance can occur in the absence of obesity. Their finding of insulin resistance as a risk factor independent of BMI is not new.

They also describe that obesity is an important factor in driving the metabolic syndrome cancer risk. Obesity is a well-known risk factor for a number of cancers.

Response

We acknowledge that previous literatures observed that insulin resistance is linked to increased risk of cancer, sometimes in the absence of obesity. In the current manuscript, we found that our machine-learning based prediction model of insulin resistance, AI-IR, outperforms BMI alone as well as previously established metrics including MetS, TG/HDL ratio and TyG index regarding the incidence of new-onset diabetes and the composite cancer. (page 20, line 340 in the revised manuscript).

“We acknowledge that the BMI-independent effect of insulin resistance on increased cancer risk is not a new finding. While the definitions of MetS, TG/HDL ratio, and TyG index do not include BMI, MetS—originally developed as an indicator of visceral adiposity—has been reported as a risk factor for certain types of cancer^{12–17}. Indeed, we observed that higher BMI (≥ 30) alone was not associated with an increased risk of composite cancer (Fig. 5a). AI-IR was a better predictor of composite cancer

compared to BMI and TyG index (Fig. 5a, 5d), but its predictive capability was comparable to that of MetS and TG/HDL ratio (Fig. 5b and 5c). Having said that, since AI-IR demonstrated the best predictive capability for the incidence of diabetes among the examined metrics, including MetS and TG/HDL ratio (Fig. 2), and diabetes itself is a significant risk factor for the composite cancers (Fig. 4a), we argue that AI-IR is the single best metric for predicting both diabetes and composite cancer incidence. Furthermore, AI-IR captures both BMI-dependent and BMI-independent effects of insulin resistance.”

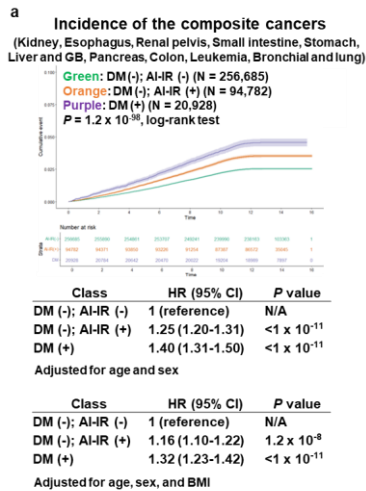


Figure 4

We also add discussion in page 21, line 355 in the revised manuscript.

“Our prediction model is based on nine commonly measured clinical parameters, and we demonstrated that it outperforms previously established simpler metrics—including BMI, MetS, TG/HDL ratio, and TyG index—in predicting both diabetes and composite cancer incidence. Our model provides “digital biomarker of insulin resistance” which merges all parameters currently established as a component of insulin resistance into a single metric, which will be utilized for identification of high-risk individuals and focused screening of diabetes, cardiovascular disease, and cancer.”

The authors trained the model in their other manuscript on the NHANES and Taiwan populations. Did they take into account that insulin resistance frequently occurs in Asian populations at lower BMI than in European populations.

Response

We included ethnicity (Asian versus non-Asian) into our prediction model when we

developed prediction model (eClinicalmedicine 2023). Therefore, our model did take into account that insulin resistance frequently occurs in Asian populations at lower BMI than in European populations. While current manuscript largely refers to non-Asian populations, we will conduct dedicated studies to examine the effect of our model, AI-IR, on the incidence of cancer in Asian populations.

In the statistical analysis, did the authors correct for multiple comparisons? Are all comparisons between all groups statistically different in Table 1, and supplementary table 1?

Response

The authors would like to thank the reviewer for this really important comment. We performed Bonferroni correction after multiple testing in the revised manuscript (Page 11, line 177 in the revised manuscript).

“However, when we looked at individual types of cancer, both AI-IR and diabetes were associated with a significantly higher risk of incidence of multiple cancer types (The results of the 25 common cancers are shown in Fig. 3, and the complete list can be found in Supplementary Table 4). We examined the incidence of 36 common cancers (both male and female), 4 cancer types specific to males, and 3 cancer types specific to females. The Bonferroni-corrected *P* value for significance was $0.05/43 = 1.163 \times 10^{-3}$.”

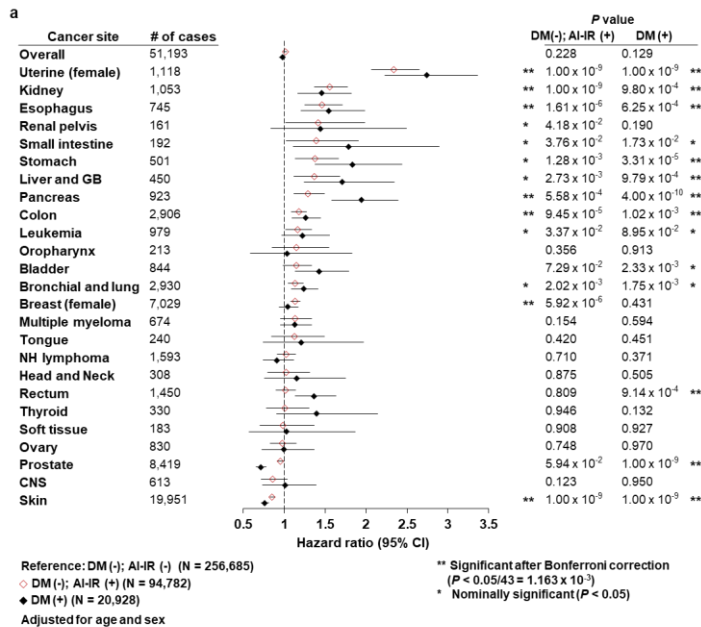


Figure 3

The results were mentioned in page 12, line 193 in the revised manuscript.

“When we considered Bonferroni correction, AI-IR was associated with a higher incidence of uterine, kidney, esophagus, pancreas, colon, and breast cancers.”

In the discussion section, beginning at line 254, the authors state that the ML model could be implemented in clinical practice at low cost. Given that we already have clinical criteria to define the metabolic syndrome, that correlates with insulin resistance, and is a well-known risk factor for diabetes and cardiovascular disease, that are inexpensive and easy to measure in any clinical practice, a critical question that the authors do not address is what the advantage of their ML model is over the current metabolic syndrome definition.

Response

We acknowledge that defining MetS is inexpensive and easy in clinical practice. Having said that, as we already mentioned in response to the previous points, the current study demonstrated that AI-IR outperforms previously established metrics including MetS in terms of the prediction of diabetes. Regarding the prediction of the incidence of the composite cancers, the capability of AI-IR was comparable to that of MetS. However, since diabetes itself is a significant risk factor for the composite cancers, collectively, we argue that AI-IR is the single best metric for predicting both diabetes and composite cancer incidence.

We revised the discussion by removing the argument regarding cost and ease of use while incorporating a comparison with previously established metrics (page 21, line 353 in the revised manuscript).

“The strength of our study is that, to our knowledge, it is the first to leverage the use of a machine learning-based prediction model of insulin resistance to look at the effect of insulin resistance on the incidence of cancer at population scale. Our prediction model is based on nine commonly measured clinical parameters, and we demonstrated that it outperforms previously established simpler metrics—including BMI, MetS, TG/HDL ratio, and TyG index—in predicting both diabetes and composite cancer incidence. Our model would provide “digital biomarker of insulin resistance” which merges all parameters currently established as a component of insulin resistance into a single metric, which will be utilized for identification of high-risk individuals and focused screening of diabetes, cardiovascular disease, and cancer.”

In the manuscript the authors cite two mendelian randomization studies looking at MR predicted fasting insulin and cancer risk. There are a number of similar MR studies, some of which are summarized in this review article.

<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC9398112/>

Response

In response to this comment, we additionally cite the indicated review article (page 19, line 319 in the revised manuscript).

“In line with our observations, a Mendelian randomization using 18 variants associated with fasting insulin levels found that genetically predicted higher fasting insulin levels increased the risk of kidney, pancreas, and lung cancer⁴. Similar Mendelian randomization studies have been replicated and summarized elsewhere³³.”

Minor comments

Line 59:

- o Current: “In this regard, it is concerned that insulin resistance and hyperinsulinemia...”
- o Suggested: “in this regard, it is theorized that” or “In this regard, it has been proposed that” etc.

Response

We modified to “In this regard, it has been proposed that...”. Thank you for your suggestion.

Line 61-62:

- o Should be “insulin insulin-like growth factor 1 receptor”

Response

We corrected.

Line 62:

- o Missing article
- o Current: “Mechanistic link between inflammation and cancer..”
- o Suggested “The mechanistic link between inflammation and cancer..”

Response

We corrected the sentence.

Line 241:

- o Current: “Future researches are anticipated to explore..”
- o Suggested: “Future research is anticipated to explore”

Response

We substantially revised the discussion section to address all the reviewers' concerns. During the revision, we removed the corresponding statement to better focus on the main scope of our manuscript, particularly the comparison of AI-IR with previously established metrics. Nevertheless, we sincerely appreciate the reviewer's careful and thorough reading of our work.

Reviewer #3 (Remarks to the Author): Expert in insulin resistance, insulin signalling, and cancer

In their manuscript, “Machine learning-predicted insulin resistance is a risk factor for 12 site-specific cancers” Lee et al use a previously described machine learning predicted insulin sensitivity score to examine the impact of insulin resistance upon cancer incidence, arguing that this score could have clinical utility for identifying patients who may benefit from increased screening. The question the authors are asking is very significant to the field because, as they highlight, it has been difficult to assess the impact of many of the individual features of systemic metabolism upon cancer risk. Unfortunately, the findings as presented need to be contextualized and interpreted in the context of both the obesity and cancer literature as well as the cancer metabolism/genetics literature, for as written the authors overstate the potential utility of this approach for cancer risk assessment and make claims related to potential causal relationships (e.g. sun exposure in the obese) for which no supportive data is provided, when activation of INSULIN signaling in certain mutational contexts drive senescent phenotypes (as a potential example in melanoma). By contrast a discussion of the mutational profiles of Uterine and Esophageal tumors both of which frequently mutate downstream components of the insulin signaling cascade (e.g. PI3K, PTEN) may help to explain why these observations are biologically significant. The key issue is that as described throughout their manuscript, the model relies heavily on BMI which is already a known risk factor for 13 tumor types (Lauby-Secretan B, 2016 NEJM, PMID:27557308). This well established observation diminishes the potential utility of the model since BMI is one of its key inputs. It might be helpful to compare how BMI alone performs as a risk factor for these tumor types, and then potentially examine the impact of insulin sensitivity scores in patients with “normal” BMIs, as that population would not already be identifiable by their physicians as being at elevated risk.

The work has significant promise, but as written it again needs to be contextualized in order for readers to be able to understand its significance, which appears to be reduced from the claims being made by the authors. In many ways as currently written this feels like a first step, where they have applied their previously developed tool to a second data set, confirming its utility, but then highlight findings that need to be contextualized. These limitations in the data as presented and the need for additional context within the obesity and cancer literature result in the authors overstating the potential utility of their Machine Learning approach and lead them to overlook, or fail to emphasize some of their findings which are perhaps more novel and may warrant deeper discussion/analysis. In particular the impact of insulin sensitivity on lung cancer where they see a signal that may well be independent of obesity and its subsequent inflammation.

Response

The authors would like to thank this reviewer for appreciating the importance of the question we are asking in the current study. We also thank specific comments below to improve our manuscript. To address concerns raised by this reviewer, for example, we directly compared AI-IR with BMI and other previously established metrics regarding the prediction of the incidence of diabetes as well as cancer. We demonstrated that AI-IR outperforms BMI and other metrics.

As this reviewer mentions, it has been difficult to assess the impact of many of the individual features of systemic metabolism upon cancer risk. Our machine learning-based prediction model of insulin resistance, AI-IR, can provide “digital biomarker of insulin resistance” which merges all parameters currently established as a component of insulin resistance into a single model. Our model includes both BMI-dependent and -independent effect of insulin resistance.

Specific points:

The key component missing in this manuscript is an open assessment of how their model compares to the use of BMI alone. Conceptually it would also benefit if the authors engaged in a frank discussion of how their Machine Learning approach fits within the extensive literature about how the components of systemic metabolism drive cancer risk in the tumor types they are highlighting.

Response

In response to this important comment, we directly compared AI-IR with BMI alone and other previously established metrics (MetS, TG/HDL ratio, and TyG index) regarding the predictive capabilities of the incidence of diabetes. We demonstrated that AI-IR is the best predictor of diabetes incidence compared to BMI, MetS, TG/HDL ratio, and TyG index (Fig. 2a and page 9, line 134 in the revised manuscript).

“To examine whether AI-IR outperforms previously established simpler metrics of obesity, metabolic syndrome, and insulin resistance, we compared the predictive capabilities of AI-IR, BMI, metabolic syndrome (MetS), TG/HDL ratio, and TyG index for the incidence of diabetes among individuals without diabetes at baseline during the follow-up period. TG/HDL ratio^{24,25} and TyG index^{26–28} has been reported as a surrogate markers of insulin resistance. Area under the curve (AUC) of the receiver operating characteristic (ROC) curve was highest for AI-IR (0.798), followed by MetS (0.748), BMI (0.721), TyG index (0.703), and Tg/HDL ratio (0.702), indicating the AI-

IR exhibits the best predictive capability among these metrics (Fig. 2a).”

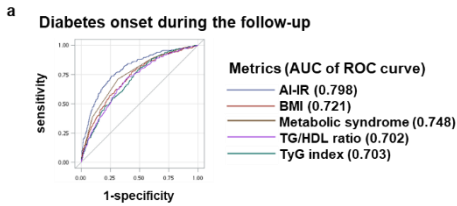


Figure 2

We also compared AI-IR and BMI regarding the prediction of the incidence of the composite cancer (page 16, line 273 in the revised manuscript).

“Finally, to examine whether AI-IR provides a better prediction of the composite cancer incidence than previously established metrics such as BMI, MetS, TG/HDL ratio, and TyG index, we classified participants who were free of diabetes and cancer at baseline into four groups based on their AI-IR status and the status of these metrics. When we consider AI-IR and BMI, we observed that the incidence of the composite cancers was significantly higher ($P = 2.28 \times 10^{-2}$) for the AI-IR single-positive group (HR, 1.18; 95% CI, 1.11-1.26; $P < 9.5 \times 10^{-8}$) compared to the BMI single-positive group (HR, 1.04; 95% CI, 0.95-1.15; $P = 0.38$). BMI of 30 or higher alone did not significantly increase the incidence of the composite cancers. We also observed that the incidence of the composite cancers was significantly higher ($P = 2.01 \times 10^{-5}$) for the AI-IR and BMI double-positive group (HR, 1.30; 95% CI 1.24-1.37; $P < 1 \times 10^{-11}$) compared to the BMI single positive group, indicating that AI-IR enables significantly improved cancer risk stratification among obese individuals (Fig. 5a).”

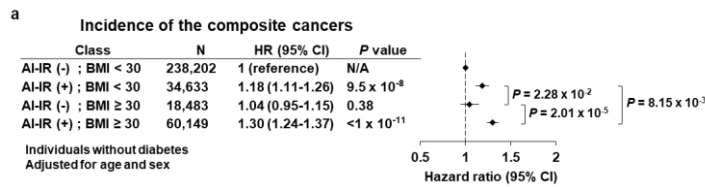


Figure 5

Also, we discussed the effect of high TG and low HDL on the incidence of lung cancer (page 20, line 332 in the revised manuscript).

“Considering that feature importance of our model was highest for higher BMI, followed by higher FPG, lower HDL, and higher TG²³, the effect of AI-IR on bronchial and lung cancer is BMI-independent but it may be dependent on FPG, HDL, and TG. Supportively, studies reported that both high TG and low HDL are positively associated with lung cancer risk^{39,40}.”

The authors fail to build out the logic that underlies their observation in the differences between smokers/non-smokers at the end of the results section. It seems as if there would be multiple variables including lung cancer screening which is typically only done in smokers and could account for the differences. Inclusion of an epidemiologist and biostatistician to ensure you are not suffering from observation bias might be useful.

Response

In response to this comment (and also in response to the comment from reviewer 1), we additionally examined the effect of AI-IR on the incidence of bronchial and lung cancer as well as composite cancers while adjusting for smoking status, which is mentioned in page 15, line 247 in the revised manuscript.

“Lung cancer is strongly associated with smoking, the leading risk factor for the global burden of cancer³¹. We observed that the effect of AI-IR on the incidence of bronchial and lung cancer is significant even when we adjusted also for smoking status (never smoker, previous smoker, and current smoker); The same is true for the incidence of the composite cancers (Supplementary Fig. 5a-b).”

a Incidence of the bronchial and lung cancer			b Incidence of the composite cancers		
Class	HR (95% CI)	P value	Class	HR (95% CI)	P value
DM (-); AI-IR (-)	1 (reference)	N/A	DM (-); AI-IR (-)	1 (reference)	N/A
DM (-); AI-IR (+)	1.19 (1.08-1.32)	5.96 x 10 ⁻⁴	DM (-); AI-IR (+)	1.12 (1.06-1.18)	2.07 x 10 ⁻⁶
DM (+)	1.25 (1.08-1.44)	2.27 x 10 ⁻³	DM (+)	1.26 (1.17-1.35)	1.30 x 10 ⁻⁹
Adjusted for age, sex, BMI, and smoking			Adjusted for age, sex, BMI, and smoking		

Supplementary Figure 5

Regarding the joint effect analysis (Supplementary Fig. 5c in the revised manuscript), we found that the incidence of lung cancer is significantly different in the same smoking status group (i.e. previous smoker), which is unlikely derived from the difference in the frequency of screening.

The claim that obesity might relate to sun exposure is not cited and would need to be backed up to remain (line 220).

Response

The authors would like to thank the reviewer for very carefully reading our manuscript. To better focus on the main scope of our manuscript, we decided to remove the claim in the revised manuscript.

The statistical methods are not robustly discussed. It would help readers interpret their findings if the authors expanded the methods to include how they are adjusting for Age, Sex,

and BMI explicitly.

Response

In response to this constructive comment, we added more detail into the method section (page 26, line 444 in the revised manuscript).

“When indicated, logistic regression and Cox proportional hazards models were adjusted for key covariates, including age, sex, and BMI, to account for potential confounding factors. To evaluate the robustness of AI-IR's effects across different covariate combinations, we conducted analyses in the following sequential steps: (1) crude (unadjusted), (2) adjusted for age and sex, and (3) adjusted for age, sex, and BMI. Age (in years) and BMI (kg/m^2) were treated as continuous variables, while sex (male or female) was included as a binary variable.”

The discussion includes a “future directions” component that discusses further validating the model, which is not the focus of this manuscript but the previous one, so the discussion of correlating hyperinsulinemic-euglycemic clamp data with HOMA-IR, and their model seems out of place.

Response

The author would like to appreciate this constructive comment. We removed our discussion on the correlation between clamp and HOMA-IR.

The key findings would significantly benefit from validation in independent data sets.

Response

The authors strongly agree with this point. However, obtaining access to independent datasets within a reasonable timeframe for revision is not feasible. We plan to conduct a dedicated study to examine the effect of AI-IR on cancer incidence in the Asian population.

Minor points:

There are several instances where the authors seem to have missed components in their editing (eg. Line 188 where it just says reference).

Response

Thank you for your comment. That “reference” in the original manuscript referred to reference point for hazard ratio. During the revision, we thoroughly reviewed and edited the

entire manuscript to enhance readability.

Methods do not include sufficient detail to evaluate study.

Response

In response to this reviewer comment, we added more description regarding the definition of diabetes in this study (page 24, line 406 in the revised manuscript), comparison of the capability of AI-IR with previously reported metrics (page 26, line 437 in the revised manuscript), and adjustment of logistic regression and Cox proportional hazards models (page 26, line 444 in the revised manuscript).

Reviewer #5 (Remarks to the Author): Early-Career Researcher co-reviewer

I co-reviewed this manuscript with one of the reviewers who provided the listed reports. This is part of the Nature Communications initiative to facilitate training in peer review and to provide appropriate recognition for Early Career Researchers who co-review manuscripts.

Response

The authors would like to appreciate your time and effort to review our manuscript.

Reviewer #6 (Remarks to the Author): Expert in machine learning in cancer

The paper “Machine learning-predicted insulin resistance is a risk factor for 12 site-specific cancers” examines the link between insulin resistance, as predicted by a machine learning model, and the risk of certain cancers. Using a machine learning model developed with clinical markers (like BMI, FPG, cholesterol levels), the study assessed insulin resistance within the UK Biobank cohort and identified associations with 12 specific cancers, such as uterine, kidney, and colon cancer. The study found that insulin resistance heightened cancer risk, with both BMI-dependent (e.g., colon cancer) and BMI-independent effects (e.g., uterine cancer).

This approach suggests that using machine learning-predicted insulin resistance could enhance targeted cancer screening and preventive strategies, especially for high-risk individuals identified through metabolic health markers. The findings of this paper could potentially have a significant impact on public health.

Response

The authors would like to the reviewer for appreciating the impact of our machine learning based prediction model of insulin resistance on public health such as targeted cancer screening and preventive strategies. In response to the specific comments below, for example, we defined composite cancers by merging 10 cancer types (common to both males and females) whose risks were either significantly or nominally increased by AI-IR and evaluated the effect of AI-IR on the incidence of the composite cancer event to enhance generalizability of our finding. Furthermore, we directly compared AI-IR with previously established simpler metrics including BMI, MetS, TG/HDL ratio, and TyG index regarding the predictive capabilities of the incidence of diabetes as well as the composite cancer.

There are a few concerns and suggestions:

1. It appears that the authors tested for insulin resistance across various cancers but found no significant association when considering all cancer types together. However, they identified 12 specific cancers (referred to as site-specific cancers) that individually show a connection to insulin resistance. While this finding is intriguing, it lacks generalizability. I suggest combining the data for these 12 site-specific cancers and re-testing the hypothesis. Since each cancer individually links to insulin resistance, a combined analysis might reveal a stronger overall signal, potentially supporting a broader association. If this additional experiment confirms a link, it could suggest an underlying biological connection across these cancers; if not, it may indicate that insulin resistance acts more as a confounder specific to these cancer types.

Response

The authors would like to thank the reviewer for this really constructive comment. This comment was very helpful to improve our manuscript and better support our conclusion. According to this reviewer comment, we defined composite cancers by merging 10 cancer types (common to both males and females) whose risks were either significantly or nominally increased by AI-IR and performed a combined analysis. The combined analysis further confirmed the link between insulin resistance and cancer. This is mentioned in page 12, line 200 in the revised manuscript.

“When we defined composite cancers by merging 10 cancer types (common to both males and females) whose risks were either significantly increased (kidney, esophagus, pancreas, colon) or nominally increased (renal pelvis, small intestine, stomach, liver and GB, colon, and bronchial and lung) by AI-IR, a Kaplan-Meier plot revealed a significant difference between participants with diabetes, those without diabetes but positive for AI-IR, and those without diabetes and negative for AI-IR, on the cumulative incidence of the composite cancers (Fig. 4a). The HR adjusted for age and sex in participants without diabetes but positive for AI-IR was 1.25 (95% CI, 1.20-1.31; $P < 1 \times 10^{-11}$), and that in participants with diabetes was 1.40 (95% CI, 1.31-1.50; $P < 1 \times 10^{-11}$).”

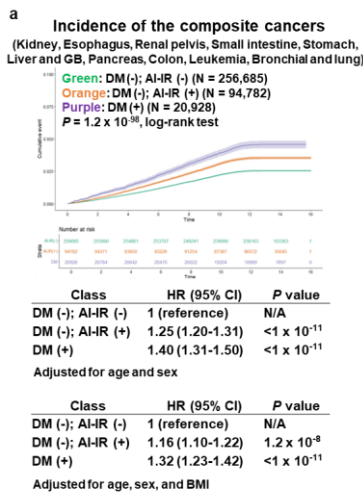


Figure 4

2. The paper’s use of a machine learning approach for predicting insulin resistance could be strengthened by comparisons to simpler, established metrics, such as the TG/HDL ratio or TG-glucose index, to evaluate whether the model offers significant advantages over these alternatives. Additionally, while the authors mention in lines 67-68 that HOMA-IR is considered the best alternative to the hyperinsulinemic-euglycemic clamp, they do not

elaborate on the reasoning behind this assertion, leaving a gap in the justification for their methodological choices.

Response

in response to this constructive comment, we directly compared AI-IR and previously reported surrogate markers of insulin resistance such as TG/HDL ratio and TyG index regarding the predictive capabilities of the incidence of diabetes. We demonstrated that AI-IR is the best predictor of diabetes incidence compared to BMI, MetS, TG/HDL ratio, and TyG index (page 9, line 134 in the revised manuscript).

“To examine whether AI-IR outperforms previously established simpler metrics of obesity, metabolic syndrome, and insulin resistance, we compared the predictive capabilities of AI-IR, BMI, metabolic syndrome (MetS), TG/HDL ratio, and TyG index for the incidence of diabetes among individuals without diabetes at baseline during the follow-up period. TG/HDL ratio^{24,25} and TyG index^{26–28} has been reported as a surrogate markers of insulin resistance. Area under the curve (AUC) of the receiver operating characteristic (ROC) curve was highest for AI-IR (0.798), followed by MetS (0.748), BMI (0.721), TyG index (0.703), and Tg/HDL ratio (0.702), indicating the AI-IR exhibits the best predictive capability among these metrics (Fig. 2a).”

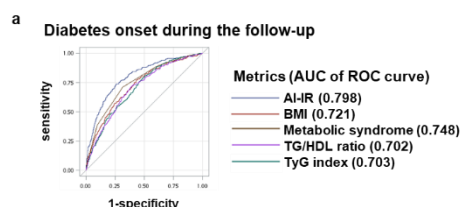


Figure 2

We also compared AI-IR with TG/HDL ratio and TyG index regarding the prediction of the incidence of the composite cancer (page 16, line 273 in the revised manuscript).

“Finally, to examine whether AI-IR provides a better prediction of the composite cancer incidence than previously established metrics such as BMI, MetS, TG/HDL ratio, and TyG index, we classified participants who were free of diabetes and cancer at baseline into four groups based on their AI-IR status and the status of these metrics.”

(page 17, line 284 in the revised manuscript)

When considering AI-IR and MetS, the AI-IR single positive group (HR, 1.21; 95% CI 1.12-1.31; $P = 2.1 \times 10^{-6}$) and the MetS single positive group (HR, 1.19; 95% CI 1.12-1.26; $P = 1.8 \times 10^{-9}$) exhibited comparable HRs ($P = 0.649$) for the incidence of the composite cancers. However, the incidence of composite cancers was significantly higher ($P = 1.49 \times 10^{-4}$) in the AI-IR and MetS double-positive group (HR, 1.34; 95%

CI 1.27-1.40; $P < 1 \times 10^{-11}$) compared to the MetS single-positive group (Fig. 5b). Similar patterns were observed when we consider AI-IR and TG/HDL ratio (Fig. 5c). When considering AI-IR and TyG index, the incidence of the composite cancers was significantly higher ($P = 1.92 \times 10^{-2}$) for the AI-IR single-positive group (HR, 1.23; 95% CI, 1.13-1.35; $P = 7.2 \times 10^{-6}$) compared to the TyG single-positive group (HR, 1.10; 95% CI, 1.05-1.16; $P = 1.1 \times 10^{-4}$). We also observed that the incidence of the composite cancers was significantly higher ($P < 1 \times 10^{-11}$) for the AI-IR and TyG double-positive group (HR, 1.32; 95% CI 1.26-1.39; $P < 1 \times 10^{-11}$) compared to the TyG single positive group (Fig. 5d). Altogether, these results indicate that AI-IR enables improved cancer risk stratification compared to previously established metrics.

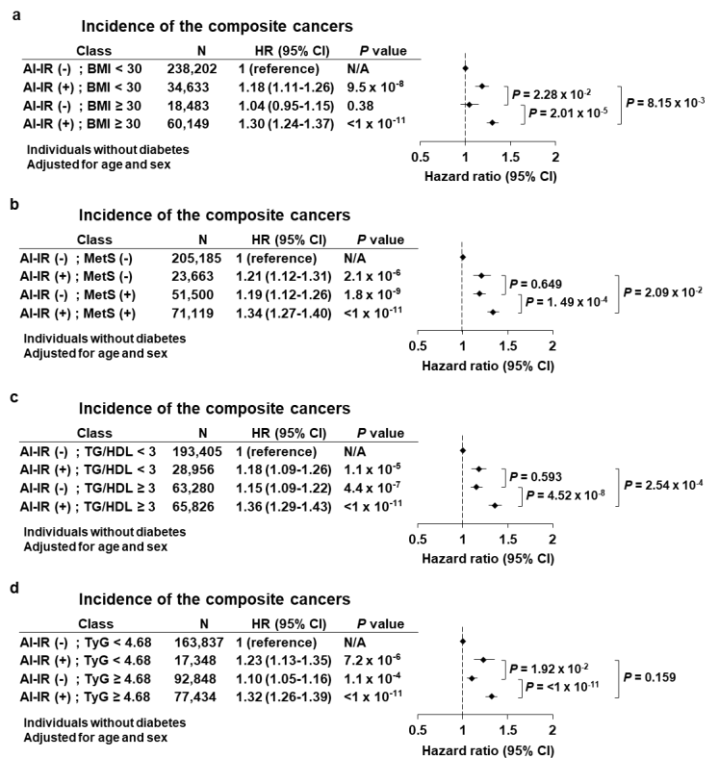


Figure 5

In the introduction, in response to this reviewer comment, we additionally explained the reason that HOMA-IR can be considered as a best alternative to the hyperinsulinemic-euglycemic clamp (page 6, line 79 in the revised manuscript).

“The homeostasis model assessment of insulin resistance (HOMA-IR) index, in which the product of fasting plasma glucose (FPG) and fasting plasma insulin is divided by a constant²⁰, has been considered a best alternative method because HOMA-IR involves the direct measurement of insulin and glucose, and have been shown to correlate strongly with the gold standard clamp technique^{21,22}.”

3. The study primarily focuses on statistical associations, without exploring the biological mechanisms that might link insulin resistance to cancer. A deeper examination of these underlying mechanisms—such as insulin’s role in cellular proliferation or its involvement in inflammation pathways—could strengthen the paper by connecting the statistical findings to established aspects of cancer biology.

Response

The authors acknowledge that investigating the biological mechanisms linking insulin resistance to cancer is an important direction for future research. Dedicated studies would be required to address this question in a thorough and meaningful way. Potential mechanisms include insulin’s role in cellular proliferation and obesity-induced inflammation. By leveraging AI-IR, for example, we can conduct genome-wide association studies on insulin resistance itself, rather than using BMI (a surrogate marker of insulin resistance) or type 2 diabetes (a consequence of insulin resistance alongside impaired insulin secretion). We added discussion in page 22, line 367 in the revised manuscript.

“Third, the mechanistic link between insulin resistance and cancer remained elusive in the current study. However, leveraging AI-IR could provide deeper insights into the biology of insulin resistance, for example, by enabling genome-wide association studies on insulin resistance itself, rather than relying on BMI (a surrogate marker of insulin resistance) or type 2 diabetes (a consequence of insulin resistance alongside impaired insulin secretion).”

4. While smoking is examined as an interaction effect in lung and bronchial cancers, it is not comprehensively addressed across other cancer types, which may influence results for cancers strongly linked to smoking, especially esophageal cancer. Including additional stratification analyses or adjusting for smoking across other cancer subtypes could help reduce potential confounding effects and enhance the robustness of the findings.

Response

In response to this comment, we performed additional analyses examining the effect of smoking along with AI-IR. First, we looked at the effect of AI-IR on the incidence of bronchial and lung cancer as well as composite cancers while adjusting for smoking status as well as age, sex, and BMI, which is mentioned in page 15, line 247 in the revised manuscript.

“Lung cancer is strongly associated with smoking, the leading risk factor for the global burden of cancer³¹. We observed that the effect of AI-IR on the incidence of bronchial

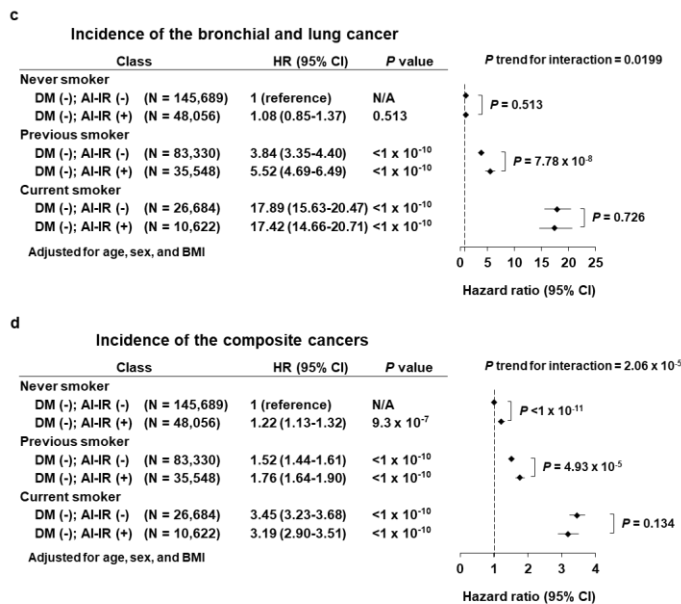
and lung cancer is significant even when we adjusted also for smoking status (never smoker, previous smoker, and current smoker); The same is true for the incidence of the composite cancers (Supplementary Fig. 5a-b)."

a Incidence of the bronchial and lung cancer			b Incidence of the composite cancers		
Class	HR (95% CI)	P value	Class	HR (95% CI)	P value
DM (-); AI-IR (-)	1 (reference)	N/A	DM (-); AI-IR (-)	1 (reference)	N/A
DM (-); AI-IR (+)	1.19 (1.08-1.32)	5.96×10^{-4}	DM (-); AI-IR (+)	1.12 (1.06-1.18)	2.07×10^{-5}
DM (+)	1.25 (1.08-1.44)	2.27×10^{-3}	DM (+)	1.26 (1.17-1.35)	1.30×10^{-9}
Adjusted for age, sex, BMI, and smoking			Adjusted for age, sex, BMI, and smoking		

Supplementary Figure 5

In addition to bronchial and lung cancer, we examined the potential joint effect of AI-IR and smoking status on the incidence of the composite cancers. We observed similar results as in the case of bronchial and lung cancer, further supporting the robustness of this finding (page 16, line 261 in the revised manuscript).

"We observed similar interaction effect on the incidence of the composite cancers; predicted insulin resistance increased the incidence in never smoker group and previous smoker group, but not in current smoker group (Supplementary Fig. 5d). Notably, the composite cancers include esophagus, pancreas, stomach, liver, and bronchial and lung cancers, all of which are known to have increased risk due to smoking³²."



Supplementary Figure 5

Point-by-point response to the editor and reviewers
Lee CL., Hiraike Y., et al. NCOMMS-24-48120A

Reviewer #2 (Remarks to the Author):

Thank you to the authors for addressing my previous comments, and editing the manuscript.

Response

The authors would like to thank the reviewer for acknowledging our effort to improve our manuscript in response to your previous comments. In light of the additional comments below, we have further revised our manuscript to improve clarity and to better support our conclusion.

Related to the metabolic syndrome compared to the AI-IR to the metabolic syndrome to predict diabetes. As age is a component of their AI-IR score, and the risk of diabetes increases with age, it is not surprising that adding age to other metabolic syndrome criteria increases the risk of diabetes. The authors adjusted for age, sex and BMI in lines 113 to 115. Did the authors adjust for age when comparing the metabolic syndrome to the AI-IR, as metabolic syndrome criteria and age are clinical parameters available to any clinician without the need for a machine learning model to predict the risk of diabetes.

Response

We thank the reviewer for this request for clarification. In our comparison of AI-IR and metabolic syndrome, we categorized individuals into four groups based on AI-IR status (positive or negative) and metabolic syndrome status (positive or negative) and assessed the incidence of diabetes (Figure 2c in the revised manuscript). We adjusted for age and sex when calculating the odds ratios. However, in the previous version of the manuscript, Figure 2b-e was mistakenly labeled as showing “univariate” analyses. We have corrected this in the revised manuscript. After adjustment for age and sex, the odds ratio for diabetes incidence was significantly higher ($P = 1.55 \times 10^{-3}$) in AI-IR single positive group (OR, 6.85; 95%CI, 4.35-10.80; $P < 1.2 \times 10^{-16}$) compared with metabolic syndrome single positive group (OR, 3.14; 95%CI, 2.07-4.77; $P < 6.8 \times 10^{-8}$).

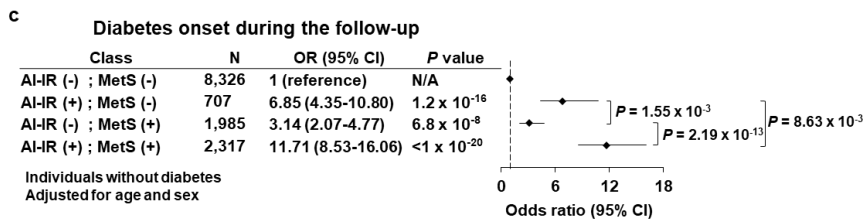


Figure 2

Lines 145 - that the AI-IR model is better than BMI to predict diabetes is not surprising, given that the AI-IR model includes the metabolic syndrome parameters that are known to be better predictors than BMI alone. It is surprising however, that the authors found no association between obesity and the risk of diabetes - do the authors have any explanation for this, as the risk of diabetes is known to increase with increasing obesity.

Response

We thank the reviewer for this important comment. In our comparison of AI-IR and BMI, individuals were categorized into four groups based on AI-IR status (positive or negative) and BMI status (≥ 30 or < 30) (Figure 2b in the revised manuscript). Thus, “BMI single positive group” represents individuals who are obese (BMI ≥ 30) but AI-IR negative. Given the parameters used to define AI-IR, these individuals are likely to have lower FPG, HbA1c, TG and higher HDL levels, which may correspond to a “metabolically healthy obesity”. We have clarified this point in page 10, line 153 in the revised manuscript.

“When we categorized individuals into four groups based on AI-IR status (positive or negative) and BMI (≥ 30 or < 30), we observed that the incidence of diabetes was significantly higher ($P = 2.52 \times 10^{-5}$) in AI-IR single-positive group (OR, 6.14; 95% CI, 4.42-8.54; $P < 1 \times 10^{-20}$; adjusted for age and sex) compared with BMI single-positive group (OR, 0.86; 95% CI, 0.35-2.12; $P = 0.74$). Of note, BMI of 30 or higher alone did not significantly increase diabetes incidence when individuals were negative for AI-IR, suggesting that these subjects may represent “metabolically healthy” obese individuals.”

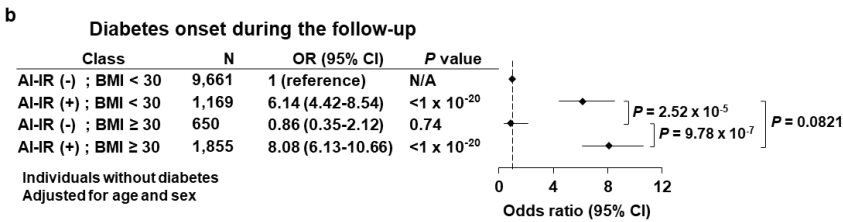


Figure 2

What is the rationale for the "double positive" groups - BMI is part of the AI-IR model, so adding BMI in again on top of that is presumably artificially inflating the association. Could the authors clarify exactly what "single positive" and "double positive" means related to BMI and metabolic syndrome. It means presence or absence of metabolic syndrome and obesity as defined by BMI?

Response

We appreciate the reviewer for this constructive comment. In our comparison of AI-IR and

BMI (Figure 2b in the revised manuscript), individuals were categorized into four groups based on AI-IR status (positive or negative) and BMI status (≥ 30 or < 30). Accordingly, “double positive group” refers to individuals who are both obese (BMI ≥ 30) and AI-IR positive. “AI-IR single positive group” represent those who are AI-IR positive but not obese (BMI < 30), and “BMI single positive group” represent those who are obese (BMI ≥ 30) but AI-IR negative. Defining groups in this way does not artificially inflate the associations. We clarified the definition of four groups in the revised manuscript.

(page 10, line 153 in the revised manuscript).

“When we categorized individuals into four groups based on AI-IR status (positive or negative) and BMI (≥ 30 or < 30),”

(page 10, line 159 in the revised manuscript).

“In contrast, diabetes incidence was markedly higher ($P = 9.78 \times 10^{-7}$) in AI-IR and BMI double-positive group (OR, 8.08; 95% CI 6.13-10.66; $P < 1 \times 10^{-20}$) compared with BMI single positive group, underscoring that AI-IR provides substantially improved diabetes risk stratification among obese (BMI ≥ 30) individuals (Fig. 2b).”

The criteria used to define the metabolic syndrome seem to be missing from the methods.

Response

We would like to appreciate this reviewer for evaluating our manuscript really carefully to improve our manuscript. In response to this comment, we have added the definition of metabolic syndrome in the Method section (page 27, line 462 in the revised manuscript).

“Metabolic syndrome (MetS) was defined as the presence of at least three of the following: (1) abdominal obesity (waist circumference ≥ 102 cm in male or ≥ 88 cm in female), (2) elevated triglycerides (≥ 150 mg/dL), (3) reduced HDL cholesterol (< 40 mg/dL in male or < 50 mg/dL in female), (4) elevated blood pressure ($\geq 130/85$ mmHg), and (5) elevated fasting glucose (≥ 100 mg/dL). Use of medications for triglycerides, HDL cholesterol, blood pressure, or glucose was not considered, as medication use was not included in the computation of AI-IR.”

In the discussion the authors argue that the AI-IR model would alter screening for diabetes, cardiovascular disease and cancer. Could the authors elaborate on what screening guidelines would be changed? The study includes individuals from the UK Biobank between 40 and 69 years old - it is unclear how would these findings would change the current screening guidelines for diabetes and cancer in individuals in this age group.

Response

We agree with the reviewer that the present study alone is not sufficient to warrant changes in existing screening guidelines for diabetes or cancer in individuals aged 40- 69 years. Our findings, however, demonstrated that AI-IR provides improved risk stratification compared with established clinical metrics such as BMI, metabolic syndrome, TG/HDL and TyG index. This suggests that, in the future, AI-IR could support identification of high-risk individuals who may benefit from earlier or more intensive monitoring. We have revised the Discussion section to clarify this point (page 22, line 379 in the revised manuscript).

“Individuals identified as AI-IR positive could undergo earlier or more frequent monitoring of glucose metabolism, cardiovascular risk factors, and age-appropriate cancer screening, thereby improving opportunities for early detection and intervention.”

The last sentence of the discussion says that the ML approach could help gain insight into disease pathophysiology. It is not clear how their AI-IR approach provides any new insight into disease pathophysiology.

Response

We appreciate the reviewer’s comment. As noted in the Discussion, we think that leveraging AI-IR could provide deeper insights into biology of insulin resistance, for example by enabling GWAS of insulin resistance. In the revised manuscript, in response to this reviewer comment, we modified the text to replace “biology of insulin resistance” with “pathophysiology of insulin resistance” to better align with the last sentence of the discussion.

(page 23, line 389 in the revised manuscript)

“However, leveraging AI-IR could provide deeper insights into the pathophysiology of insulin resistance, for example, by enabling genome-wide association studies on insulin resistance itself, rather than relying on BMI (a surrogate marker of insulin resistance) or type 2 diabetes (a consequence of insulin resistance alongside impaired insulin secretion).”

(page 23, line 395 in the revised manuscript)

“Our model exhibited the highest predictive capability for the incidence of diabetes and composite cancers among the examined metrics, highlighting the potential of a machine learning-assisted approach to enhance clinical practice and gain insights into disease pathophysiology.”

Reviewer #3 (Remarks to the Author):

The authors have been largely responsive to the concerns raised in my initial review.

Response

The authors sincerely thank this reviewer for positive evaluation of our manuscript and for acknowledging the improvements made in response to previous comments.

Reviewer #7 (Remarks to the Author): Expert in cancer epidemiology and statistics; replaces Reviewer #1

Thank you for asking me to review this paper. It is a very interesting paper with highly pertinent results. I do have a few comments for the authors:

Response

The authors sincerely thank this reviewer for positively evaluating our manuscript, as well as for the constructive and valuable comments that help us further improve our manuscript. In response to the specific comments below, we performed additional analyses, including Cox proportional hazards models using age as the underlying time scale and stratified analyses based on fasting duration prior to blood collection. We also extensively revised the Method section to enhance clarity, reproducibility, and transparency. Overall, we believe that all the concerns raised by the reviewer have been thoroughly addressed.

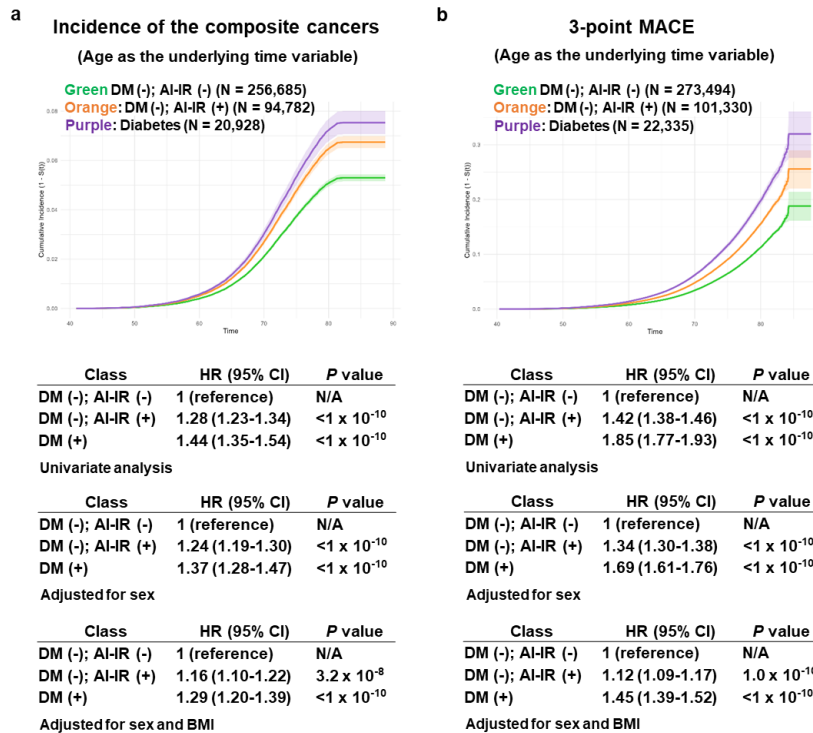
I agree with the previous reviewer, the models should have age as the underlying time variable. Adjusting and stratifying by age at recruitment is not sufficient to fully account for age in the models. To ensure your models are robust, you should re-run them with age as the underlying time variable.

Response

The authors would like to thank the reviewer for this really important comment. To account for the strong effect of age on cancer incidence, we re-ran the Cox proportional hazards models using age as the underlying time scale. This approach ensures that each participant contributes person-time from their age at enrollment until their age at censoring or event, providing a more effective control for confounding by age than including age as a covariate. We observed AI-IR remained a robust predictor of cancer incidence under this model. This is mentioned in page 14, line 238 in the revised manuscript.

“To more rigorously account for confounding by age, we also conducted analyses with age as the underlying time variable. Significant differences in cancer incidence persisted among participants with diabetes, those without diabetes but AI-IR positive, and those without diabetes and AI-IR negative (Supplementary Fig. 6a.) The HR adjusted for sex and BMI in participants without diabetes but AI-IR positive was 1.16 (95% CI, 1.10-1.22; $P < 3.2 \times 10^{-8}$), and that in participants with diabetes was 1.29 (95% CI, 1.20-1.39; $P < 1 \times 10^{-10}$). The effect of AI-IR on 3-point MACE was also consistent in these models (Supplementary Fig. 6b). Together, these findings demonstrate the robustness of AI-IR in predicting complications of insulin resistance

while fully accounting for the effect of age.”



Supplementary Figure 6

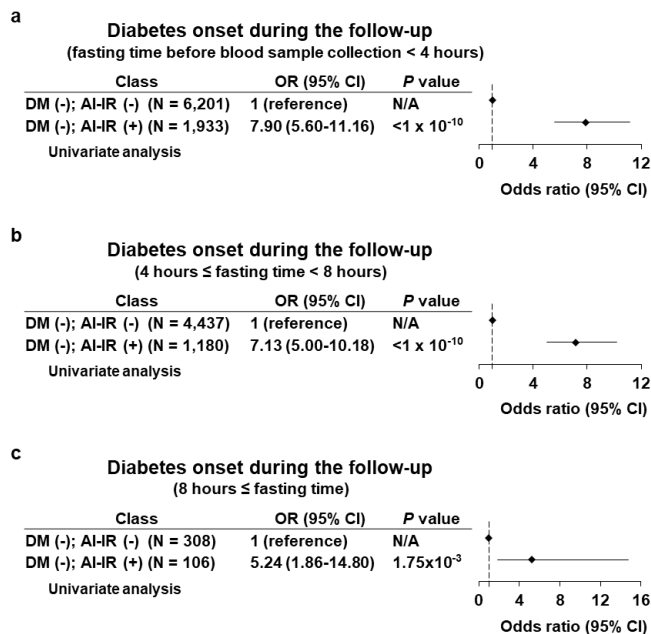
UKB does not have fasting glucose available, fasting samples were not collected. You have not addressed this in your paper, how could this affect the prediction of IR given the original model was training on fasting glucose levels? This should be addressed.

Response

We thank the reviewer for this really constructive comment. To address the concern regarding fasting status, we performed stratified analyses based on fasting duration prior to blood collection, which is described in page 7, line 107 in the revised manuscript.

“Notably, although AI-IR was originally developed using fasting blood test results, blood samples in the UK Biobank were not consistently collected in the fasted state. To address this, we stratified participants into three groups based on fasting duration prior to blood collection (less than 4 hours, 4 or more and less than 8 hours, 8 hours or more) and examined the association between AI-IR and the incidence of diabetes. Across all categories, AI-IR positive participants consistently showed a significantly higher risk of developing diabetes, supporting the robustness of our model (Supplementary Fig. 1a-c). Hereafter, to maintain maximum sample size, we did not

stratify or exclude participants based on fasting duration.”



Supplementary Figure 1

You have undertaken a impressive amount of analysis. However, this makes the methods very difficult to follow, and I'm not sure that your work would be replicable with the current methods section. I think you should consider rewriting this section so it is easy to follow and understand exactly what was undertaken in each analysis. Additionally, I am not convinced that the correct censor date has been applied to the non-cancer participants. The two data fields listed for applying censor date are country of birth data fields, so I cannot see how you used these. Further, all first cancer diagnoses are present in the cancer registry, so absence indicates no cancer diagnosis, so the censor date of the cancer registry should be applied to all participants, regardless of if they were diagnosed with a cancer or not.

Response

We thank this reviewer for this constructive comment. In response, we have extensively revised the Method section to improve clarity, reproducibility, and transparency of our analyses.

Regarding the censor date for non-cancer participants, we have also extensively revised the description and provided additional explanation (page 25, line 423 in the revised manuscript).

“Follow-up was conducted until the last available date in the registries: 31 December 2020 for England, 30 November 2021 for Scotland, and 31 December 2016 for Wales. Participants without cancer were censored at the last available date of the respective

registry, ensuring consistent follow-up for all participants. For participants recorded in the combined “Originating from England/Wales (E/W)” registry, it was not possible to assign registry-specific censor dates to each participant. In these cases, censor dates were assigned as 31 December 2016 for participants born in Wales (determined by Data-Fields 1647 and 20115), and 31 December 2020 for all others.”

We acknowledge that, for participants in the combined E/W registry, registry-specific censor dates cannot be assigned with certainty, which represents a limitation of the available registry data. As clarified by UK Biobank, Welsh cancer registry data had ceased to be included in NHS England cancer registry extracts from 2017 onwards, and the censoring date for Welsh records has been revised accordingly.

(https://biobank.ndph.ox.ac.uk/ukb/exinfo.cgi?src=Data_providers_and_dates)

How model adjustments were applied is not clear - I was under the understanding from the methods that all models were adjusted for age, sex and BMI, however the section from line 232 suggests that is not the case.

Response

We thank the reviewer for this request for clarification. Analyses were performed depending on the context: as crude (unadjusted), adjusted for age and sex, or adjusted for age, sex, and BMI, to account for potential confounding. We have carefully reviewed the entire manuscript and confirmed that model adjustments are now clearly described in the main text, figure legends, and captions.

Did you consider co-linearity between the various measures investigated, given that all these variables are included in the AI-IR model?

Response

We thank the reviewer for this insightful comment. In traditional regression models, co-linearity between predictors can bias coefficient estimates and inflate standard errors, which is a major concern. However, in machine learning models like XGboost (which we used for AI-IR), the algorithm can naturally handle correlations among input features by modeling non-linear interactions and making splits based on feature importance rather than relying on a single coefficient per variable. Therefore, correlations among input parameters do not affect the predictive performance of AI-IR.

line141 - Can you test to see if there is a significant difference between AUCs?

Response

We thank the reviewer for this suggestion. We found that AI-IR had a significantly higher AUC compared to other metrics (page 9, line 146 in the revised manuscript).

“Area under the curve (AUC) of the receiver operating characteristic (ROC) curve was highest for AI-IR (0.798, $P < 1 \times 10^{-4}$ vs other metrics), followed by MetS (0.748), BMI (0.721), TyG index (0.703), and Tg/HDL ratio (0.702), indicating the AI-IR exhibits the best predictive capability among these metrics (Fig. 2a).”

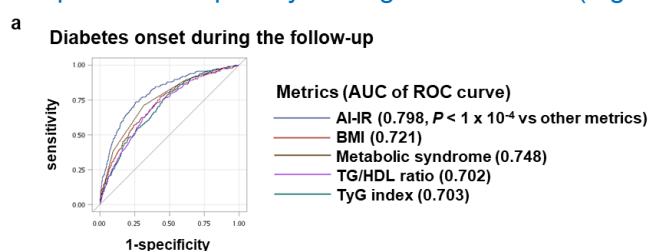


Figure 2

lines 142 - 144: it is really not clear what has been done here

Response

In response to this reviewer comment, we have revised the text to clarify our approach in Figure 2b-e. (page 10, line 149 in the revised manuscript).

“Next, we categorized individuals without diabetes at baseline into four groups according to both AI-IR status (positive or negative) and their status on previously established metrics (positive or negative).”

For example, when using BMI, individuals were categorized into four groups: (1) “double negative group” (AI-IR negative and BMI < 30), which served as the reference group; (2) “AI-IR single positive group” (AI-IR positive and BMI < 30); (3) “BMI single positive group” (AI-IR negative and BMI ≥ 30); (4) “double positive group” (AI-IR positive and BMI ≥ 30). This classification scheme was applied analogously to other established metrics.

line 148: given this contradicts previous research, can you be confident in your results? What models did you use to undertake this analysis?

Response

We thank the reviewer for this important comment. In our comparison of AI-IR and BMI

(Figure 2b in the revised manuscript), individuals were categorized into four groups based on AI-IR status (positive or negative) and BMI status (≥ 30 or < 30). Thus, “BMI single positive group” represents individuals who are obese (BMI ≥ 30) but AI-IR negative. Given the parameters used to define AI-IR, these individuals are likely to have lower FPG, HbA1c, TG and higher HDL levels, which may correspond to a “metabolically healthy obesity”. We have clarified this point in page 10, line 153 in the revised manuscript.

“When we categorized individuals into four groups based on AI-IR status (positive or negative) and BMI (≥ 30 or < 30), we observed that the incidence of diabetes was significantly higher ($P = 2.52 \times 10^{-5}$) in AI-IR single-positive group (OR, 6.14; 95% CI, 4.42-8.54; $P < 1 \times 10^{-20}$; adjusted for age and sex) compared with BMI single-positive group (OR, 0.86; 95% CI, 0.35-2.12; $P = 0.74$). Of note, BMI of 30 or higher alone did not significantly increase diabetes incidence when individuals were negative for AI-IR, suggesting that these subjects may represent “metabolically healthy” obese individuals.”

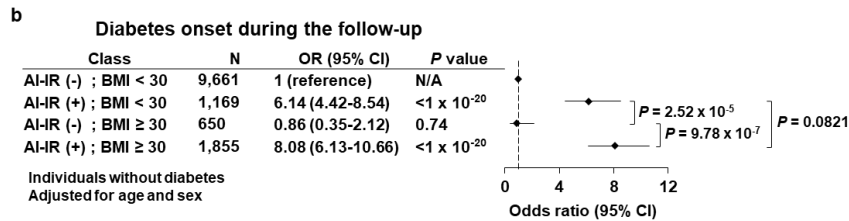


Figure 2

lines 153 - again, it's really unclear what you have done here, what are the 4 groups?

Response

In response to this comment, we clarified the explanation to define four groups for Figure 2c (page 10, line 163 in the revised manuscript).

“Similarly, when we categorized individuals without diabetes into four groups based on both AI-IR status (positive or negative) and MetS (positive or negative)”

Individuals were categorized into four groups: (1) “double negative group” (AI-IR negative and MetS negative), which served as the reference group; (2) “AI-IR single positive group” (AI-IR positive and MetS negative); (3) “MetS single positive group” (AI-IR negative and MetS positive); (4) “double positive group” (AI-IR positive and MetS positive).

lines 184-193: I think only Bonferroni sig results should be reported

Response

The authors would like to thank the reviewer for this valuable comment. In the previous revision, reviewer #6 (who has been replaced by reviewer #8 in the current round) recommended that, to increase generalizability, we define “composite cancers” by combining cancer sites showing either significant or nominal associations with AI-IR. Following this suggestion, we constructed the composite cancer outcome and performed pooled analyses. Therefore, we consider it important to report both significant and nominal associations—including those for renal pelvis, small intestine stomach, liver and GB, leukemia, and bronchial and lung—as they provide transparency and clarify how the composite cancer was defined.

lines 200 - 217: I'm not convinced this section adds to the paper, it feels like analysis for analysis sake, rather than truly trying to understand the aetiology (which would be best done individually for each cancer site)

Response

We thank the reviewer for this comment. Related to the previous point, in response to reviewer #6 in the previous round, we defined “composite cancers” to increase generalizability of our findings. We believe this composite approach adds value in two important ways: (1) it increases statistical power by pooling less-common cancers, thereby reducing the risk of false negatives; and (2) it provides a more clinically relevant and generalizable picture of the overall cancer burden associated with AI-IR, which would be overlooked if we only examined site-specific outcomes. While detailed investigation of individual cancer sites is certainly important for understanding aetiology, the composite cancer analysis serves a different purpose—offering a broader assessment of overall cancer risk associated with AI-IR.

line 404: there is no non-UK area covered by UKB - can you clarify what you mean by this?

Response

We appreciate the reviewer’s request for clarification. The phrase “non-UK area” was introduced in reference to country of birth (Data-Fields 1647 and 20115), which was misleading since UK Biobank only includes participants residing in the UK. In the revised manuscript, we corrected this point and extensively revised the Method section to clarify how these data-fields were used when assigning censor dates for participants without cancer (page 25, line 423 in the revised manuscript).

“Follow-up was conducted until the last available date in the registries: 31 December 2020 for England, 30 November 2021 for Scotland, and 31 December 2016 for Wales. Participants without cancer were censored at the last available date of the respective registry, ensuring consistent follow-up for all participants. For participants recorded in the combined “Originating from England/Wales (E/W)” registry, it was not possible to assign registry-specific censor dates to each participant. In these cases, censor dates were assigned as 31 December 2016 for participants born in Wales (determined by Data-Fields 1647 and 20115), and 31 December 2020 for all others.”

line 410: include ICD codes used.

Response

In response to this reviewer comment, we have added the ICD codes used to define diabetes (page 25, line 434 in the revised manuscript).

“while long-term follow-up relied on NHS-recorded diagnosis (ICD-10 E08, 09, 11, and 13), with diagnosis date defined as the date of admission with diabetes.”

Reviewer #7 (Remarks on code availability):

Whilst the code for the machine learning predicted insulin resistance is under patent, I do feel like the code for this analysis should be provided (that is, the code for these analysis, without the code to generate the AI-IR values) so it is transparent exactly how the analysis was undertaken.

Response

We thank the reviewer for this important comment. To ensure transparency while protecting our filed patent, we consulted our patent lawyer, who strongly advised against disclosing the source code prior to the publication of the patent (please see the documents “250805_Case Status_NAIPO.pdf” and “250805_Case Status_NAUPPO.pdf” uploaded as a related manuscript files). Consequently, the code for computing AI-IR from the nine input parameters cannot be publicly released at this time.

To provide transparency while respecting patent restrictions, we have created a website that allows users to calculate AI-IR based on the nine input parameters. The current website is managed on a local server, but we plan to deploy it on a public cloud server to ensure broader accessibility upon publication of this work.

<http://220.130.208.108:8000>

Reviewer #8 (Remarks to the Author): Expert in machine learning, epidemiology, and omics; replaces Reviewer #6

Authors seemed to have addressed Reviewer #6 comments.

Response

The authors sincerely thank the reviewer for acknowledging our efforts to address the comments raised by Reviewer #6 and for recognizing the improvements made to our manuscript.

Point-by-point response to the editor and reviewers

Lee CL., Hiraike Y., et al. NCOMMS-24-48120B

Reviewer #2 (Remarks to the Author):

EDITORIAL NOTE: This referee was not able to submit their report on this occasion. However, Reviewer #3 assessed the response. While they consider that most points were addressed, they noted that, when comparing between BMI-high and AI-IR positive groups, there is still some concern regarding the underlying issue that BMI may account for diabetes risk in a manner that considerably overlaps with AI-IR prediction.

Response

The authors would like to thank the Reviewer #3 for evaluating our response and for positively acknowledging our efforts to address the points raised in the previous review.

Regarding the potential overlap between BMI and AI-IR in terms of diabetes prediction, in response to the EDITORIAL NOTE, we added discussion, which is mentioned in page 19, line 327 in the revised manuscript.

“We acknowledge that BMI is the most influential single parameter for AI-IR. When we categorized individuals into four groups based on AI-IR status (positive or negative) and BMI (≥ 30 or < 30) (Fig. 2b), BMI may contribute to diabetes risk in a manner that partially overlaps with AI-IR. Nevertheless, AI-IR significantly outperformed BMI and other metrics in predicting diabetes incidence, consistent with the notion that AI-IR captures multidimensional clinical and biochemical abnormalities beyond BMI alone.”

Reviewer #7 (Remarks to the Author):

Thank you for resubmitting this manuscript. You have addressed all my previous concerns.

Response

The authors sincerely thank the reviewer for acknowledging our efforts to address the previous concerns.